MSc DATA SCIENCE - CST4090 INDIVIDUAL PROJECT

USING MACHINE LEARNING TO PREDICT AND ANALYSE CYCLONE EVENTS: A DATA_DRIVEN APPROACH TO TROPICAL STORM FORECASTING

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ABSTRACT

Tropical cyclones (TCs) are among the most destructive meteorological hazards, often causing severe socio-economic disruption, large-scale displacement of communities, and considerable loss of life (Needham et al., 2015; Emanuel, 2017). Despite advances in meteorological forecasting systems, accurately predicting the intensity, trajectory, and rapid intensification of cyclones remains challenging, particularly for small island developing states (SIDS) and low-lying coastal nations, where vulnerability to storm surges and flooding is disproportionately high (Camargo & Wing, 2016; Knutson et al., 2020). As climate change continues to amplify cyclone risks through warming sea surface temperatures (SSTs) and greater atmospheric moisture (IPCC, 2021), the need for robust and adaptive forecasting tools has become both a scientific imperative and a humanitarian necessity.

This dissertation addresses these gaps by applying machine learning (ML) methods to enhance cyclone forecasting, integrating storm track data from the International Best Track Archive for Climate Stewardship (IBTrACS) (Knapp et al., 2010) with atmospheric and oceanic predictors from ERA5, the fifth-generation reanalysis produced by the European Centre for Medium-Range Weather Forecasts (Hersbach et al., 2020). Variables including mean sea level pressure (MSLP), SST, and 10 m wind components (u- and v-winds) were selected due to their established meteorological relevance.

The research adopts a quantitative, positivist methodology, combining statistical hypothesis testing with machine learning benchmarks. Three hypotheses were tested: (i) longer storm duration is associated with higher maximum wind speeds, (ii) average storm latitude is weakly related to intensity, and (iii) lower central pressure strongly correlates with higher wind speeds. Results confirmed Hypotheses 1 and 3, while Hypothesis 2 was only weakly supported.

A series of ML and DL models—Random Forest, XGBoost, LightGBM, Support Vector Regression (SVR), Long Short-Term Memory (LSTM) networks, and Temporal Convolutional Networks (TCN)—were developed and compared using RMSE, MAE, and R^2. LightGBM achieved the best performance (RMSE = 7.72, MAE = 5.23, R^2 = 0.92), outperforming both traditional methods and deep learning models. Feature importance analysis consistently identified central pressure as the strongest predictor, reinforcing long-established meteorological theory.

The findings provide three major contributions: empirical benchmarking of ML and DL approaches for cyclone intensity prediction, validation of cyclone–climate relationships using statistical and ML interpretability methods, and the development of a reproducible ML pipeline with direct implications for disaster preparedness and climate adaptation strategies. Limitations include the restricted temporal scope (2019–2020) and the underperformance of DL models due to dataset size, pointing to future work incorporating larger datasets, additional predictors (e.g., wind shear, humidity), and hybrid ML–DL ensembles.

This study demonstrates the potential of machine learning—particularly gradient boosting models—in improving cyclone forecasting skill, offering both scientific insight and practical value for climate resilience and policy.

Keywords: *Tropical Cyclones (TCs)***, *Machine*** *Learning (ML), climate forecasting, Sea Surface Temperature (SST), ERA5 Reanalysis, IBTrACS, disaster preparedness, climate change adaptation, Random Forest, Long Short-Term Memory (LSTM), XGBoost.*

List of Abbreviations

1. **TCs** – Tropical Cyclones
2. **SIDS** – Small Island Developing States
3. **UNDRR** – United Nations Office for Disaster Risk Reduction
4. **IPCC** – Intergovernmental Panel on Climate Change
5. **SST** – Sea Surface Temperature
6. **NWP** – Numerical Weather Prediction
7. **NOAA** – National Oceanic and Atmospheric Administration
8. **GFS** – Global Forecast System
9. **ECMWF** – European Centre for Medium-Range Weather Forecasts
10. **IFS** – Integrated Forecasting System
11. **ML** – Machine Learning
12. **DL** – Deep Learning
13. **ERA5** – Fifth Generation ECMWF Atmospheric Reanalysis
14. **IBTrACS** – International Best Track Archive for Climate Stewardship
15. **RI** – Rapid Intensification
16. **LSTM** – Long Short-Term Memory
17. **CNNs** – Convolutional Neural Networks
18. **TCWCs** – Tropical Cyclone Warning Centres
19. **RSMCs** – Regional Specialized Meteorological Centres
20. **MLP** – Mean Sea Level Pressure
21. **SVR** – Support Vector Regression
22. **TC** – Tropical Cyclone
23. **CNN** – Convolutional Neural Network
24. **RNN** – Recurrent Neural Network
25. **TCN** – Temporal Convolutional Network
26. **EPS** – Ensemble Prediction System
27. **NCEP** – National Centers for Environmental Prediction
28. **WMO** – World Meteorological Organization
29. **RI** – Rapid Intensification
30. **SHIPS** – Statistical Hurricane Intensity Prediction Scheme
31. **XAI** – Explainable Artificial Intelligence
32. **SHAP** – SHapley Additive exPlanations
33. **RBF** – Radial Basis Function (kernel used in SVMs)
34. **RF** – Random Forest
35. **XGBoost** – Extreme Gradient Boosting
36. **LightGBM** – Light Gradient Boosting Machine
37. **SVM** – Support Vector Machine
38. **LIME** – Local Interpretable Model-agnostic Explanations
39. **ANFIS** – Adaptive Neuro-Fuzzy Inference System
40. **SLT** – Statistical Learning Theory

41. **RQ** – Research Question
42. **GPU** – Graphics Processing Unit
43. **CDS** — Copernicus Data Store
44. **API** — Application Programming Interface
45. **NetCDF** — Network Common Data Form
46. **ConvLSTM** — Convolutional Long Short-Term Memory
47. **GAN / GANs** — Generative Adversarial Network(s)
48. **FMM** — Finite Mixture Model
49. **SURGEDAT** — Global Storm Surge Database (SURGEDAT)
50. **RMSE** — Root Mean Squared Error
51. **MAE** — Mean Absolute Error
52. **R^2** — Coefficient of Determination
53. **EDA** – Exploratory Data Analysis
54. **SP** – South Pacific
55. **WP** – Western Pacific
56. **NI** – North Indian Ocean
57. **SI** – South Indian Ocean
58. **EP** – Eastern Pacific
59. **ITCZ** – Intertropical Convergence Zone
60. **RF** – Random Forest
61. **SVR** – Support Vector Regression
62. **CNN** – Convolutional Neural Network
63. **GAN** – Generative Adversarial Network
64. **FMM** – Finite Mixture Model
65. **R^2** – Coefficient of Determination
66. **XAI** – Explainable Artificial Intelligence
67. **SHAP** – SHapley Additive exPlanations
68. **H_0** – Null Hypothesis
69. **H_1** – Alternative Hypothesis
70. **r** – Pearson Correlation Coefficient
71. **ρ** – Spearman Rank Correlation Coefficient
72. **p** – p-value (statistical significance)
73. **ENSO** – El Niño–Southern Oscillation
74. **IOD** – Indian Ocean Dipole
75. **OHC** – Ocean Heat Content
76. **SHAP** – SHapley Additive Explanations
77. **LIME** – Local Interpretable Model-agnostic Explanations
78. **AWS** – Amazon Web Services
79. **Kaggle** – Online platform for data science and machine learning competitions
80. **SVR** – Support Vector Regression

Table of Content

1.0 INTRODUCTION 1

1.1 BACKGROUND AND GLOBAL CONTEXT 1
1.2 FORECASTING APPROACHES: FROM NUMERICAL MODELS TO MACHINE LEARNING 2
1.3 DATA FOUNDATIONS FOR ML-BASED CYCLONE PREDICTION 3
1.4 RESEARCH GAP 4
1.5 PROBLEM STATEMENT 5
1.6 RATIONALE 5
1.7 RESEARCH AIM AND OBJECTIVES 6
1.8 RESEARCH CONTRIBUTIONS 7
1.9 RESEARCH SIGNIFICANCE 8

2.0 LITERATURE REVIEW 9

2.1 TROPICAL CYCLONE (TC) FORECASTING RESEARCH 9
2.2 FORECASTING FOUNDATIONS 10
2.2.1 Lifecycle Theory of Cyclones 10
2.2.2 Coriolis Effect and Latitude Constraints 10
2.2.3 Pressure–Wind Relationship 10
2.3 TRADITIONAL FORECASTING APPROACHES: NWP/SHIPS/ENSEMBLE FORECASTING 11
2.3.1 Numerical Weather Prediction (NWP) 11
2.3.2 Statistical–Dynamical Models 11
2.3.3 Ensemble Forecasting 11
2.3.4 Limitations and Constraints 12
2.4 DEEP LEARNING FOR CLIMATE AND WEATHER FORECASTING 13
2.4.1 Recurrent Neural Networks (RNNs) and LSTM 14
2.4.2 Temporal Convolutional Networks (TCN) 14
2.4.3 CNNs and Spatial Modelling 14
2.4.4 Transformers in Weather Forecasting 14
2.4.5 Limitations & Constraints of Deep Learning Models 15
2.5 MACHINE LEARNING THEORIES 16
2.5.1 Statistical Learning Theory (SLT) 16
2.5.2 Ensemble Learning Theory 16
2.5.3 Kernel Theory in SVMs 17
2.5.4 Sequence Learning Theory in LSTM 17
2.5.5 Representation Learning in Convolutional Neural Networks (CNNs) and 17
Temporal Convolutional Networks (TCNs) 17
2.5.6 Interpretability and Explainable AI (XAI) 17
2.6 CLASSICAL MACHINE LEARNING METHODS IN METEOROLOGY 18
2.7 FEATURE ENGINEERING IN METEOROLOGICAL ML 18
2.8 ADVANTAGES AND LIMITATIONS OF ML MODELS 19
2.9 Challenges in Data-Driven Cyclone Forecasting 20
2.9.1 Data Sparsity and Quality 20
2.9.2 Interpretability and Trust 20
2.9.3 Computational Constraints 21
2.9.4 Integration with Existing Systems 21
2.10 USE OF IBTRACS AND ERA5 IN TC FORECASTING RESEARCH 21
2.11 CONSTRAINTS AND LIMITATIONS 22

2.12 HYPOTHESES FORMULATION 23
2.13 SUMMARY 25

3.0 METHODOLOGY 26

3.1 RESEARCH DESIGN 26
3.1.1 Research Philosophy 28
3.1.2 Research Approach 28
3.1.3 Methodological Choice 28
3.1.4 Research Strategy 29
3.1.5 Time Horizon 30
3.2 DATA SOURCES 30
3.2.1 IBTrACS Dataset 30
3.2.2 ERA5 Reanalysis Dataset 31
3.3 DATA PREPROCESSING 31
3.3.1 IBTrACS Cleaning 31
3.3.2 ERA5 Cleaning 32
3.3.3 Data Integration 32
3.3.4 Feature Engineering 32
3.3.5 Scaling and Normalisation 33
3.3.6 Hypothesis Testing 33
3.4 DATA PREPARATION AND FEATURE ENGINEERING 33
3.5 MACHINE LEARNING MODELS 34
3.5.1 Classical ML Models 34
3.5.2 Deep Learning Models 35
3.6 MODEL VALIDATION AND EVALUATION METRICS 35
3.7 INTERPRETABILITY AND OPERATIONAL APPLICABILITY 35
3.8 IMPLEMENTATION ENVIRONMENT 36
3.9 RATIONALE AND REFLECTION 36
3.10 SUMMARY 36

4.0 RESULTS AND ANALYSIS 37

4.1 EXPLORATORY DATA ANALYSIS (EDA) 37
4.1.1 Wind Speed by Basin 37
4.1.2 Distribution of Sea Surface Temperature (SST) 39
4.1.3 Distribution of Central Pressure 40
4.1.4 Distribution of Wind Speed 41
4.1.5 Geographical Distribution of Cyclone Tracks 42
4.1.6 Integrative Insights 43
4.2 OVERVIEW OF EXPERIMENTAL RESULTS 43
4.3 PERFORMANCE OF CLASSICAL MACHINE LEARNING MODELS 46
4.3.1 Random Forest 46
4.3.2 XGBoost 47
4.3.3 LightGBM 47
4.3.4 Support Vector Regression 47
4.4 PERFORMANCE OF DEEP LEARNING MODELS 48
4.4.1 LSTM 48
4.4.2 Temporal Convolutional Network (TCN) 48
4.5 COMPARATIVE PERFORMANCE ANALYSIS 49
4.6 ERROR ANALYSIS 49
4.7 WHY CLASSICAL ML BEAT DEEP LEARNING 50

4.8 ERROR PATTERN INSIGHTS 50
4.9 PRACTICAL IMPLICATIONS 50
4.10 IMPLICATIONS 51

5.0 HYPOTHESES AND DISCUSSION 52

5.1 RATIONALE FOR HYPOTHESIS SELECTION 52
5.2 RESULTS AND INTERPRETATION OF HYPOTHESES 54
5.3 OVERALL ASSESSMENT 59
5.4 LINKING HYPOTHESES TO MODEL RESULTS 60
5.5 IMPLICATIONS 61

6.0 LIMITATIONS AND FUTURE WORK 62

6.1 INTRODUCTION 62
6.2 LIMITATIONS OF THE STUDY 62
6.2.1 Dataset Scope and Size 62
6.2.2 Feature Availability and Resolution 62
6.2.3 Model Performance and Deep Learning Underperformance 63
6.2.4 Interpretability Concerns 64
6.2.5 Generalisation and External Validity 64
6.2.6 Hypothesis Testing Limitations 64
6.3 FUTURE WORK 65
6.3.1 Expanding Dataset Scope 65
6.3.2 Enhancing Feature Engineering 65
6.3.3 Deep Learning Advancements 65
6.3.4 Model Interpretability 66
6.3.5 Computational Improvements 66
6.3.6 Expanding Evaluation Frameworks 66
6.4 CONCLUSION 67

7.0 CONCLUSION 68

7.1 OVERVIEW OF THE STUDY 68
7.2 KEY FINDINGS 68
7.3 CONTRIBUTIONS OF THE STUDY 69
7.4 LIMITATIONS REVISITED 70
7.5 FUTURE DIRECTIONS 70
7.6 FINAL REFLECTIONS 71

REFERENCES 72

List of Table

Table 1: Comparison of Traditional Cyclone Forecasting Approaches.....13
Table 2: Limitations & Constraints of DL.....15
Table 3: Advantages & Limitations of ML in Meteorology.....19
Table 4: Analytical Mapping of Hypotheses to Theory.....25
Table 5 : Results.....44

List of Figure

Figure 1: The Research Onion Framework (Saunders et al., 2003) 27
Figure 2: Research design workflow illustrating the positivist, quantitative, and machine learning-driven approach applied in this study......27
Figure 3: Wind Speed by Basin 37
Figure 4:Distribution of SST 39
Figure 5: Distribution Central Pressure (hPa) 40
Figure 6: Distribution of Wind Speed.....41
Figure 7: Scatterplot -Geographical Distribution of Cyclone Tracks.....42
Figure 8 : RMSE Model Comparison 45
Figure 9: Model Comparison 45
Figure 10: Model Comparison - R^2 values 46
Figure 11: Results and Interpretation of Hypotheses.....54
Figure 12: Scatterplot H1 55
Figure 13: H2 57
Figure 14 : H3 58

1.0 INTRODUCTION

1.1 Background and Global Context

Tropical cyclones (TCs)known regionally as hurricanes or typhoons—are among the most destructive natural hazards worldwide, exerting profound humanitarian and economic impacts on a global scale. These storms are characterised by extreme winds, torrential rainfall, storm surges, and flash floods that devastate infrastructure, economies, and communities (Ma, 2014; Emanuel, 2018). They frequently disrupt critical infrastructure, displace populations, and lead to significant loss of life and livelihoods. Globally, TCs cause billions of dollars in damages annually, with disproportionate effects on vulnerable populations in Small Island Developing States (SIDS), South Asia, Africa, and low-lying coastal regions (UNDRR, 2019; IPCC, 2021; Vousdoukas et al., 2023).

Historical records highlight the catastrophic humanitarian consequences of inadequate forecasting. The 1970 Bhola Cyclone struck present-day Bangladesh, claiming over 300,000 lives, making it the deadliest storm in recorded history (Frank & Husain, 1971). Typhoon Haiyan (2013) devastated the Philippines, generating storm surges exceeding seven metres, resulting in more than 6,000 fatalities (Needham et al., 2015). More recently, Cyclone Idai (2019) struck Mozambique, Zimbabwe, and Malawi, displacing millions and causing over 1,300 deaths (UNDRR, 2019). Cyclone Amphan (2020) inflicted USD 13 billion in damages across India and Bangladesh, ranking among the costliest storms in South Asia's history (Mahmud & Prowse, 2021). These events underscore the high stakes of cyclone forecasting—not only as a scientific challenge but as a humanitarian necessity.

In the context of climate change, the risks associated with TCs are intensifying. Rising sea surface temperatures (SSTs), shifting circulation patterns, and increasing atmospheric moisture are influencing storm frequency, intensity, and duration, with growing evidence of more frequent rapid intensification (Knutson et al., 2020; IPCC, 2021). Accurate and timely forecasting is therefore of immense societal importance.

National meteorological services and global agencies rely primarily on Numerical Weather Prediction (NWP) systems, such as NOAA's Global Forecast System (GFS) and the ECMWF Integrated Forecasting System (IFS). These models simulate atmospheric processes by solving physical equations of motion, with skill depending heavily on initial conditions and parameterisation of physical processes (Bauer, Thorpe & Brunet, 2015). While NWP remains the operational gold standard, limitations remain: these models are computationally intensive, sensitive to initial-condition errors, and often struggle to represent fine-scale TC structure and rapid intensity changes. Forecast errors—especially in predicting rapid intensification or unusual tracks—continue to challenge disaster preparedness and response (DeMaria et al., 2014; Emanuel, 2016; Oguejiofor et al., 2023).

The emergence of machine learning (ML) and deep learning (DL) has introduced a new paradigm in meteorological research. Unlike NWP, ML models can uncover nonlinear patterns in large datasets without requiring explicit physical representation. This data-driven approach has shown promise in complementing existing tools, especially when applied to high-resolution reanalysis datasets such as ERA5 (Hersbach et al., 2020) and observational archives such as IBTrACS (Knapp et al., 2010). By leveraging both environmental predictors (e.g., SST, wind components, pressure) and cyclone-specific features (e.g., track, duration, intensity), ML/DL approaches have demonstrated potential in providing more adaptive and accurate forecasts (McGovern et al., 2019; Chen, Gopalakrishnan & Marks, 2019; Giffard-Roisin et al., 2020; Accarino et al., 2023).

1.2 Forecasting Approaches: From Numerical Models to Machine Learning

National meteorological services and global agencies rely primarily on Numerical Weather Prediction (NWP) models such as the ECMWF Integrated Forecasting System (IFS) and NOAA's Global Forecast System (GFS). These physics-based systems simulate atmospheric conditions by numerically solving conservation laws and the equations of motion. Forecast skill depends critically on the accuracy of initial conditions and parameterisation of unresolved processes (Bauer, Thorpe & Brunet, 2015).

While NWP remains the operational benchmark, significant challenges persist; models are computationally intensive, sensitive to initial-state errors, and often struggle to represent fine-scale structure and rapid intensity changes of TCs (DeMaria et al., 2014; Emanuel, 2016, 2018; Oguejiofor et al., 2023). Forecast errors, especially in predicting rapid intensification (RI) and unusual storm tracks—continue to create operational risks for disaster preparedness and response (Camargo & Wing, 2016).

In response, machine learning (ML) and deep learning (DL) approaches have emerged as promising complements to physics-based models. Instead of explicitly solving physical equations, ML techniques identify nonlinear patterns in large reanalysis and observational datasets, enabling improved capture of cyclone-related variability (McGovern et al., 2019). Applications include cyclone genesis classification using decision-tree ensembles and support vector machines (Chen, Gopalakrishnan & Marks, 2019), sequential track modelling with Long Short-Term Memory (LSTM) networks (Kim et al., 2019), and spatio-temporal intensity prediction using Convolutional Neural Networks (CNNs) and Temporal Convolutional Networks (Hu et al., 2017; Bai, Kolter & Koltun, 2018). ML has also been successfully applied to satellite imagery (Zhang et al., 2017), radar data (Chen, Wu & Duan, 2020; Huang et al., 2021), and surface observations (Mercer & Grimes, 2015).

Despite this promise, ML applications remain fragmented: most are basin-specific, lack systematic benchmarking, and offer limited interpretability—barriers to operational integration (Molnar, 2020; Hossain, Rafiei & Choudhury, 2022).

1.3 Data Foundations for ML-Based Cyclone Prediction

The effectiveness of ML-based forecasting depends on robust, globally representative datasets:
IBTrACS (International Best Track Archive for Climate Stewardship): NOAA's unified archive of tropical cyclone data, providing six-hourly storm tracks, maximum winds, and central pressure, harmonized across international agencies (Knapp et al., 2010). For this study, 2019–2024 records are extracted, yielding ~22,000 observations across 494 storms.
ERA5 Reanalysis: The fifth-generation ECMWF reanalysis dataset, offering global coverage of key atmospheric variables such as mean sea-level pressure (MSLP), sea surface temperature (SST), and wind vectors at 0.25° spatial resolution and hourly intervals (Hersbach et al., 2020).

Preprocessing aligns IBTrACS with ERA5 temporally and spatially, including interpolation and feature engineering (e.g., SST anomalies, wind magnitude), ensuring consistency for ML model training.

1.4 Research Gap

Although numerical weather prediction (NWP) models have advanced track forecasting, their skill in predicting cyclone intensity and rapid intensification (RI) remains constrained by sparse ocean observations, difficulties in modelling nonlinear air–sea interactions, and computational demands (Bauer et al., 2015; Emanuel, 2016; DeMaria et al., 2014). In parallel, machine learning (ML) and deep learning (DL) approaches have demonstrated strong potential by uncovering hidden, nonlinear relationships in reanalysis and observational datasets (McGovern et al., 2019; Giffard-Roisin et al., 2020).

However, critical limitations persist in existing ML/DL research:

1. Basin-specific focus – most studies test models on a single ocean basin, limiting global generalisability (Zhang et al., 2017; Chen et al., 2019).
2. Fragmented benchmarking – few studies conduct systematic, comparative evaluations across multiple ML and DL algorithms to identify the most reliable frameworks (Hossain et al., 2022).
3. Weak theoretical integration – many models lack grounding in established meteorological principles (e.g., thermodynamic theory of cyclone intensity, pressure–wind relationship), reducing scientific validity.
4. Operational limitations – limited interpretability and cross-basin validation hinder real-world adoption by forecasting centres (Molnar, 2020; IPCC, 2021).

This reveals a clear research gap: the absence of a globally representative, theory-informed, and systematically benchmarked evaluation of ML/DL models that can balance accuracy, generalisability, and operational utility.

1.5 Problem Statement

Despite advances in both physics-based and data-driven methods, forecasting tropical cyclone intensity and evolution remain a major challenge as numerical weather prediction models are hindered by sparse ocean observations, computational demands, and difficulties in representing complex atmospheric interactions. While machine learning and deep learning offer potential to capture nonlinear relationships across multi-source datasets, most existing studies are basin-specific, lack systematic benchmarking, and provide limited interpretability, restricting operational uptake. These limitations leave intensity forecasts significantly less reliable than track forecasts, creating major risks for preparedness and response (Emanuel, 2016; Oguejiofor et al., 2023).

This study addresses these gaps by specifically evaluating the potential of both classical ML models (Random Forest, XGBoost, LightGBM, and Support Vector Regression) and deep learning models (Long Short-Term Memory networks and Temporal Convolutional Networks). The study integrates multi-source datasets—IBTrACS and ERA5 reanalysis to train, test, and compare models on cyclone-related predictions.

The central problem tackled is whether ML models, when trained on relevant cyclone and environmental features, can achieve reliable forecasting accuracy and thereby complement or improve upon existing forecasting approaches

1.6 Rationale

Tropical cyclone forecasting remains constrained by persistent challenges, especially in predicting intensity and rapid intensification (RI). Despite advances in numerical weather prediction (NWP), models often struggle to capture nonlinear air–sea interactions and high-resolution vortex dynamics, leading to errors with serious consequences for evacuation planning and disaster preparedness (Montgomery et al., 2009; Halperin et al., 2013; Wang, 2018). These shortcomings are most critical in vulnerable regions such as the Bay of Bengal, where surges above 10 metres have caused repeated humanitarian and economic crises (Frank & Husain, 1971; Needham et al., 2013).

Machine learning (ML) provides a powerful alternative, with demonstrated capacity to model hidden nonlinear relationships across high-dimensional data. Studies show that deep learning models (e.g., CNNs, LSTMs) significantly improve track and intensity prediction, while ensemble approaches (e.g., Random Forest, XGBoost, LightGBM) enhance reliability in cyclone detection and intensification forecasting (Ahijevych et al., 2016; Chen et al., 2019; Wimmers et al., 2019; Lee et al., 2020; Zhang et al., 2019). Yet, existing research remains fragmented, basin-specific, and inconsistently benchmarked, limiting operational uptake (Molnar, 2020; Hossain et al., 2022).

This study addresses those gaps by systematically benchmarking classical and deep learning models using globally integrated datasets (IBTrACS, and ERA5). In doing so, it advances both the scientific understanding of cyclone behaviour and the operational capacity of forecasting systems.

Ultimately, the research is justified on two grounds: scientifically, it strengthens hybrid approaches that complement physics-based models; societally, it aligns with the urgent need to improve early warning and climate resilience in cyclone-prone regions (IPCC, 2021). By integrating theory testing with model training, the study ensures that its findings are both scientifically grounded and operationally relevant.

1.7 Research Aim and Objectives

The overall aim of this dissertation is to evaluate the effectiveness of machine learning and deep learning techniques for improving the prediction and analysis of tropical cyclone events.

The aim has been broken into specific measurable objectives which are:

1. To collect, pre-process, and integrate IBTrACS and ERA5 datasets for robust hypotheses testing and model training.
2. To develop and implement multiple ML models, including Random Forest, XGBoost, LightGBM, Support Vector Machines, and deep learning models such as LSTM and TCN.

3. Test a climate hypothesis using the third dataset, specifically investigating the statistical relationships between cyclone intensity and environmental factors.
4. To statistically examine relationships between cyclone intensity and key predictors using Pearson's and Spearman's correlations, validating theoretical models of TC behaviour.
5. To Evaluate and interpret model performance using appropriate statistical metrics and explainability tools to assess predictive accuracy and practical utility

1.8 Research Contributions

This research makes both theoretical and practical contributions:

Theoretical Contributions:

- Demonstrates the application of advanced ML/DL methods to TC forecasting.
- Provides a systematic comparative analysis of classical ML versus DL architectures.
- Validates the use of globally integrated datasets (IBTrACS and ERA5) to bridge meteorology and data science.

Practical Contributions:

- Offers methodological insights into feature engineering, preprocessing, and dataset integration for meteorological ML pipelines.
- Produces comparative results that can guide operational agencies in adopting hybrid forecasting systems.
- Identifies key predictors (e.g., minimum central pressure, storm duration, SST) that can be prioritised in real-time operational forecasting.

By bridging meteorology and data science, this research not only addresses existing gaps in forecasting but also informs operational agencies seeking to adopt hybrid systems, thereby strengthening climate resilience, and laying a foundation for future interdisciplinary research.

1.9 Research Significance

This research is significant in three interrelated respects.

i. It advances academic knowledge by addressing the core aim of the dissertation—developing and evaluating ML/DL models for cyclone prediction—through systematic data integration *(Objective 1)* and comparative model development *(Objective 2)*.

ii. It contributes to practice by testing theory-driven hypotheses *(Objective 3)* and offering robust, replicable methods for meteorological agencies, thereby bridging statistical analysis with operational forecasting.

iii. It supports policy by providing empirical insights into cyclone–climate relationships and model performance *(Objective 4),* delivering evidence relevant to disaster preparedness and climate adaptation strategies, particularly in vulnerable regions.

2.0 Literature Review

2.1 Tropical Cyclone (TC) Forecasting Research

Tropical cyclone forecasting has fundamentally relied on physics-based methods, notably Numerical Weather Prediction (NWP), which solve the governing atmospheric equations to simulate storm dynamics (Courtney et al., 2019). Despite their central role in operational meteorology, NWP models still struggle with accurately forecasting intensity and rapid intensification, particularly because of challenges in vortex initialization and convective parameterization (Courtney et al., 2019; Zhang et al., 2023).

These limitations have spurred data-driven approaches such as machine learning (ML) and deep learning (DL) that can learn complex nonlinear relationships from data without explicit physical parameterisation (Mohanty et al., 2020; Xu et al., 2022). For instance, DL models using multilayer perceptrons have outperformed traditional statistical–dynamical schemes in 24-h intensity prediction, reducing forecast errors by 5 %–22 % (Xu et al., 2021). Recent hybrid CNN models have further demonstrated improvements over operational forecasts for 24- to 72-h lead times, especially in rapid intensification cases (Lee et al., 2024)

This chapter critically reviews the literature across four domains:

1. Traditional physics-based cyclone forecasting methods.
2. Machine learning methods applied to meteorology.
3. Deep learning models tailored for sequential and spatiotemporal prediction.
4. Cross-cutting challenges such as interpretability, data quality, and computational scalability.

The review identifies both the progress and research gaps that motivate this dissertation's hypotheses and modelling strategy.

2.2 Forecasting Foundations

Forecasting foundations in TC research rest on two interlinked traditions:

- physics-based numerical weather prediction, which operationalises atmospheric dynamics (NWP systems)
- data-driven approaches, which identify empirical patterns to address persistent gaps in track and intensity forecasting (ML, DL, and hybrid frameworks).

2.2.1 Lifecycle Theory of Cyclones

TCs evolve through distinct phases—genesis, intensification, maturity, and decay—each governed by complex thermodynamic and dynamic interactions (Emanuel, 2005). Storm duration is a critical factor within this lifecycle, as longer-lived systems have greater opportunity to interact with favourable oceanic and atmospheric conditions, such as elevated sea surface temperatures (SSTs) and reduced vertical wind shear. These sustained interactions increase the likelihood of intensification, including episodes of rapid intensification (RI) (Camargo & Wing, 2016).

This theoretical foundation directly informs *H1: Duration vs. Maximum Wind Speed*

2.2.2 Coriolis Effect and Latitude Constraints

Gray's (1975) theory of necessary conditions for cyclone genesis emphasises latitude as a threshold factor. Without sufficient Coriolis force (~5° from the equator), cyclones cannot develop. Beyond genesis, latitude continues to influence storm behaviour through its modulation of SST gradients, shear, and poleward moisture transport (Emanuel, 2017).

This leads to *H2: (Latitude vs. Wind Speed),* to test whether latitude's influence extends beyond formation into peak intensity.

2.2.3 Pressure–Wind Relationship

The inverse correlation between minimum central pressure and maximum sustained winds is a cornerstone of cyclone dynamics. Theoretical models (Emanuel, 2017) and empirical conversions (Knaff & Zehr, 2007) consistently validate this principle, operationalised in

intensity estimation techniques such as Dvorak. This provides direct theoretical justification for *H3 (Pressure vs. Wind Speed).*

2.3 Traditional Forecasting Approaches: NWP/SHIPS/Ensemble Forecasting

2.3.1 Numerical Weather Prediction (NWP)

Numerical Weather Prediction (NWP) models simulate atmospheric conditions by solving the primitive equations of fluid dynamics and thermodynamics. Prominent systems such as the ECMWF Integrated Forecasting System (IFS) and NOAA's Global Forecast System (GFS) have substantially improved tropical cyclone (TC) track forecasts, with track errors reduced by 60–70% over recent decades due to advances in model resolution, physical parameterisation, and data assimilation (Bauer, Thorpe & Brunet, 2015). However, intensity prediction remains significantly less reliable, particularly during episodes of rapid intensification (RI). Persistent limitations include poor vortex initialization, uncertainties in air–sea coupling, and inadequate representation of convective processes (Montgomery et al., 2009; Emanuel, 2016; Knutson et al., 2020).

2.3.2 Statistical–Dynamical Models

To mitigate some of these shortcomings, statistical–dynamical approaches have been developed, combining regression-based methods with environmental predictors derived from dynamical models. The Statistical Hurricane Intensity Prediction Scheme (SHIPS) remains the most widely used example, incorporating predictors such as sea surface temperature (SST), vertical wind shear, humidity, and persistence (DeMaria & Kaplan, 1994; DeMaria et al., 2005). While SHIPS consistently improves short-term intensity guidance relative to climatology and persistence, its reliance on manually engineered predictors and linear assumptions constrains performance during nonlinear phenomena such as RI (DeMaria et al., 2014; Wang et al., 2023). Recent comparisons highlight these shortcomings: Wimmers, Velden & Cossuth (2019) showed that LSTM-based deep learning models reduced RI forecast errors by ~20% compared to SHIPS, while Lee et al. (2024) demonstrated that hybrid deep learning approaches substantially outperformed SHIPS in both RI and peak intensity forecasts.

2.3.3 Ensemble Forecasting

Ensemble forecasting improves probabilistic skills by running multiple simulations with perturbed initial conditions, thereby quantifying forecast uncertainty. Global ensemble

prediction systems (EPS) such as those run by ECMWF and NCEP have enhanced track prediction reliability and operational decision-making (Lu et al., 2024). However, ensemble means tend to underestimate extremes in intensity, including RI events, largely because they inherit structural limitations of the underlying dynamical models (Zhang & Weng, 2015). Moreover, ensemble systems require substantial computational resources, limiting their operational application in resource-constrained meteorological services (WMO, 2012; WMO, 2019).

2.3.4 Limitations and Constraints

Despite marked progress, traditional forecasting approaches continue to face systemic constraints.

- NWP models provide physics-based realism and strong track forecast skill but struggle with intensity prediction due to initialization, convective-scale errors, and insufficient coupling (Montgomery et al., 2009; Emanuel, 2016; Knutson et al., 2020).
- Statistical–dynamical models such as SHIPS improve over persistence baselines but remain constrained by their linear structures and limited ability to capture nonlinear intensification dynamics (Wang et al., 2023; Lee et al., 2024).
- Ensemble methods enhance uncertainty quantification and probabilistic track skill but remain computationally expensive and prone to underestimating intensity extremes (Lu et al., 2024; WMO, 2019).

These limitations collectively explain the persistent forecast skill gap in rapid intensification and highlight the growing need for data-driven alternatives such as machine learning (ML) and deep learning (DL), which offer greater capacity to capture nonlinear, multiscale processes.

Table 1: Comparison of Traditional Cyclone Forecasting Approaches

Approach	Strengths	Limitations/Constraints
Numerical Weather Prediction (NWP)	Physics-based realism; global coverage; track forecast errors reduced by 60–70% over past decades due to advances in resolution and data assimilation.	Weak intensity and RI prediction due to poor storm initialization, unresolved convective processes, and limited ocean–atmosphere coupling.
Statistical–Dynamical Models (e.g., SHIPS)	Incorporates environmental predictors (SST, shear, humidity, persistence); improved short-term intensity forecasts relative to climatology and persistence.	Constrained by linear assumptions and manually engineered predictors; underperforms in RI events compared to ML/DL methods.
Ensemble Forecasting	Provides probabilistic guidance by sampling forecast uncertainty; improves track skill and enhances operational decision-making.	Computationally expensive; ensemble means underestimate intensity extremes; inherits dynamical model deficiencies.

2.4 Deep Learning for Climate and Weather Forecasting

Deep learning (DL) has become a transformative tool in meteorology, enabling models to capture complex nonlinear spatio-temporal dependencies that traditional physics-based or statistical frameworks often miss. Building on breakthroughs in computer vision and sequence learning (LeCun, Bengio & Hinton, 2015; Goodfellow, Bengio & Courville, 2016), DL methods have been applied to tropical cyclone prediction, intensity estimation, and global forecasting. Architectures such as Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformers increasingly complement Numerical Weather Prediction (NWP), enhancing accuracy and computational efficiency (Rasp, Dueben & Scher, 2020; Lam et al., 2022).

2.4.1 Recurrent Neural Networks (RNNs) and LSTM

Long Short-Term Memory (LSTM) networks (Hochreiter & Schmidhuber, 1997) are designed to overcome vanishing gradient issues in sequential learning. In meteorology, they capture temporal dependencies in cyclone lifecycle data, such as storm evolution and rapid intensification phases. Applications to typhoon and hurricane track prediction show improved accuracy over classical regression and shallow ML models, particularly when large datasets are available (Kim & Kim, 2021). However, their performance remains sensitive to preprocessing choices and gaps in observational data.

2.4.2 Temporal Convolutional Networks (TCN)

Temporal Convolutional Networks (TCN) (Bai, Kolter & Koltun, 2018) extend convolutional methods to sequence modelling using dilated causal convolutions. Unlike RNNs, TCNs allow efficient parallel training and capture long-range temporal dependencies without recurrent connections. Although still emerging in meteorological applications, early studies suggest TCNs can outperform LSTMs in medium-range forecasting and rainfall prediction tasks (Sønderby et al., 2020; Lim et al., 2021). Their scalability makes them promise for cyclone forecasting, though empirical adoption is still limited.

2.4.3 CNNs and Spatial Modelling

Convolutional Neural Networks (CNNs) excel at extracting spatial patterns, making them highly suited to satellite imagery and reanalysis datasets. They have been applied to cyclone detection, intensity classification, and wind field estimation (Chen et al., 2019; Pradhan et al., 2018). By integrating CNNs with sequential models such as LSTMs, hybrid architectures can exploit both spatial and temporal dependencies, enhancing cyclone intensity forecasting (Kim et al., 2021). Nevertheless, CNNs require large, annotated datasets, which remain a constraint for rare, high-intensity events.

2.4.4 Transformers in Weather Forecasting

Transformers (Vaswani et al., 2017), originally developed for natural language processing, leverage attention mechanisms to model long-range dependencies without recurrence. Their adaptation to weather forecasting has shown promising results: Lam et al. (2022) demonstrated

Transformer-based models outperforming traditional NWP in global medium-range forecasts, while Pathak et al. (2022) introduced FourCastNet, a Fourier Neural Operator–Transformer hybrid achieving state-of-the-art skill in climate simulations. Despite their success, Transformers are computationally demanding, raising feasibility concerns for operational use in resource-limited meteorological services.

2.4.5 Limitations & Constraints of Deep Learning Models

Table 2: Limitations & Constraints of DL

Model Type	Strengths	Limitations / Constraints
LSTM (RNNs)	Captures long-term temporal dependencies; effective for cyclone lifecycle modelling.	Sensitive to missing data and preprocessing; struggles with exceptionally long sequences; risk of overfitting small datasets.
TCN	Efficient training; avoids vanishing gradients; handles long-range dependencies well.	Still emerging in meteorology; less interpretability; requires careful tuning of dilation/filters.
CNN	Extracts spatial patterns from reanalysis and satellite imagery; enhances hybrid frameworks.	Limited in capturing temporal evolution alone; prone to ignoring storm lifecycle dynamics.
Transformers	Models global spatio-temporal dependencies; scalable to large datasets; competitive accuracy.	Extremely computationally intensive; requires large training datasets; difficult to deploy in real-time operations.
General DL Constraints	Flexible, scalable, capable of nonlinear learning; potential to outperform traditional methods.	"Black-box" nature reduces interpretability; computationally expensive; data sparsity (especially RI events) limits reliability.

2.5 Machine Learning Theories

Unlike physics-based numerical models that depend on explicit atmospheric equations, machine learning (ML) theories emphasise statistical learning and data-driven generalisation. By capturing nonlinear, high-dimensional patterns in large datasets, ML offers predictive flexibility where traditional frameworks struggle. These theories provide the foundation for testing this study's hypotheses: sequence learning frameworks operationalise storm duration (H1), kernel methods reveal nonlinear latitude–intensity links (H2), and ensemble and representation learning approaches strengthen the pressure–wind relationship (H3). In this way, ML theories not only complement meteorological science but also extend it through scalable, adaptive forecasting strategies.

2.5.1 Statistical Learning Theory (SLT)

Vapnik (1995) SLT provides the foundation for supervised learning, generalisation bounds, and feature selection. The bias–variance trade-off (Geman et al., 1992) explains why linear regression underfits nonlinear cyclone–environment interactions, while deep models risk overfitting sparse cyclone datasets. Pearson and Spearman correlations serve as statistical pre-tests of feature relevance (storm duration, latitude, pressure), aligning with SLT principles of empirical risk minimisation.

2.5.2 Ensemble Learning Theory

Breiman's (2001) Random Forests and Friedman's (2001) boosting theory show that aggregating weak learners reduces variance and improves robustness.

Ensemble forecasting theory holds that perturbing initial conditions yields probabilistic guidance that better represents uncertainty (Palmer, 2019). Yet, while ensemble means improve track skill, they continue to underestimate extreme intensification events (Zhang & Weng, 2015). This reinforces the need for alternative methods that better capture nonlinear extremes. Ensemble ML models such as RF, XGBoost, and LightGBM have shown superior skill in intensity forecasting compared to regression baselines (Zhang et al., 2019).

Their ability to capture nonlinear storm–environment interactions directly support H1 and H3.

2.5.3 Kernel Theory in SVMs

Cortes & Vapnik (1995) demonstrated that kernel methods allow nonlinear separation in higher-dimensional spaces. Applied to cyclone data, SVM kernels (e.g., Radial Basis Function (RBF)) can capture subtle nonlinearities between latitude and storm intensity that linear models miss. The RBF kernel allows SVMs to capture hidden, curved relationships — for instance, how latitude subtly affects maximum wind speed. This provides a methodological basis for testing H2.

2.5.4 Sequence Learning Theory in LSTM

Hochreiter & Schmidhuber (1997) introduced Long Short-Term Memory (LSTM) networks to overcome vanishing gradients in sequential learning. In meteorology, LSTMs model cyclone lifecycle evolution, enabling temporal features like storm duration to be operationalised (Rüttgers et al., 2019). This directly operationalises H1.

2.5.5 Representation Learning in Convolutional Neural Networks (CNNs) and Temporal Convolutional Networks (TCNs)

Representation learning (LeCun et al., 2015) enables models to automatically extract relevant spatio-temporal features. CNN's have been applied to satellite imagery for intensity estimation (Chen et al., 2019), while TCNs extend convolutional frameworks to sequential tasks, avoiding vanishing gradient problems (Bai et al., 2018). CNN's and TCNs extend deep learning into spatio-temporal cyclone prediction. CNN's extract spatial patterns from satellite and reanalysis data, critical for understanding intensity and pressure–wind dynamics (H3), while TCNs adapt convolution to time-series, effectively modelling storm lifecycles (H1) and latitude-linked dynamics (H2). Together, they represent advanced representation learning frameworks capable of operationalising cyclone science in ML forecasting These models are theoretically well-suited to testing spatio-temporal hypotheses (H2, H3).

2.5.6 Interpretability and Explainable AI (XAI)

One persistent barrier to ML adoption is its black-box nature. Explainable AI frameworks (Ribeiro et al., 2016; Molnar, 2020) emphasise transparent feature attribution through methods such as SHAP values and saliency maps. For cyclone forecasting, XAI validates whether storm

duration, latitude, and central pressure emerge as dominant features, linking ML models back to meteorological theory.

2.6 Classical Machine Learning Methods in Meteorology

Building on these theories, classical ML models have been adapted for meteorological prediction. Random Forests (RF) demonstrate robustness and interpretability, with studies showing moderate but consistent improvements in genesis prediction relative to logistic regression (Zhang et al., 2019). XGBoost and LightGBM, both gradient-boosted tree ensembles, excel in handling high-dimensional meteorological predictors, efficiently addressing missing data and nonlinear interactions (Chen & Guestrin, 2016). Meanwhile, Support Vector Regression (SVR) has been effective for high-dimensional but small-sample cyclone data, particularly in wind speed and surge prediction (Mercer & Grimes, 2015). Comparative studies highlight their superiority over regression-based baselines, though scalability and interpretability remain constraints (Pan, 2019).

2.7 Feature Engineering in Meteorological ML

While ML theories enable nonlinear, high-dimensional learning, feature engineering operationalises meteorological science into structured predictors. Cyclone features drawn from IBTrACS include storm duration, latitude, minimum central pressure, and maximum sustained winds—directly aligned with this study's hypotheses (Knapp et al., 2010). These are complemented by ERA5 reanalysis variables such as SST anomalies, mean sea level pressure, vertical wind shear, and humidity (Hersbach et al., 2020). Recent studies show that including dynamic environmental predictors alongside static climatological means significantly improves ML intensity forecasts (Chen et al., 2019; Lee et al., 2020). However, preprocessing challenges persist, particularly in aligning multi-basin datasets with inconsistent reporting standards (Wang, 2018).

Feature engineering thus serves as the bridge between theoretical constructs (e.g., duration as a proxy for intensification potential, latitude as Coriolis forcing, pressure as structural intensity)

and their empirical operationalisation within ML/DL frameworks. In this sense, feature design is not merely technical but theoretically informed, ensuring that hypotheses remain grounded in cyclone science rather than arbitrary data-driven discovery.

2.8 Advantages and Limitations of ML Models

Table 3: Advantages & Limitations of ML in Meteorology

Category	Advantages	Limitations / Constraints
High-dimensional modeling	ML models excel in managing large, complex datasets and extracting subtle patterns (e.g., in many meteorological variables) (Zhang et al., 2025)	The "curse of dimensionality" means overfitting is a risk without proper regularisation or dimensionality reduction (Bonavita, M. (2020); Bonetti, P., Metelli, A. M., & Restelli, M. (2023).
Nonlinear relationships	ML can learn complex nonlinear links that linear statistical models cannot (de Burgh-Day & Leeuwenburg, 2023)	Without domain constraints, ML may learn spurious correlations, especially in out-of-sample or extreme cases (Bonavita et al., 2024)
Computational efficiency at inference	After training, ML models often make predictions much faster than full NWP runs (de Burgh-Day & Leeuwenburg, 2023)	Training and hyperparameter tuning remain computationally expensive; models may not generalise well across domains
Limited temporal / sequential modelling	Standard ML models (e.g., RF, boosting) do not inherently capture temporal dependencies	Capturing storm evolution over time requires sequence models (RNN, LSTM) or feature engineering
Data sensitivity	ML requires high-quality and representative training data; noise, bias, missing values degrade performance	For rare events or extremes, training samples are inherently limited (Watson, 2022)
Black-box behaviour / interpretability	Some ML models offer variable importance, but deeper models are opaque	Lack of transparency is a hurdle for operational adoption; interpretability remains an active research area (Yang et al., 2024)

2.9 Challenges in Data-Driven Cyclone Forecasting

Despite progress in machine learning, applying data-driven methods to tropical cyclone forecasting remains challenging. Grounded in statistical learning theory (Vapnik, 1995) and limited by data quality and availability (Kossin et al., 2013; Hersbach et al., 2020), key obstacles include data sparsity, model interpretability, computational demands, and integration with physics-based numerical weather prediction (NWP). Overcoming these constraints is essential for ML to complement, rather than replace, traditional forecasting systems (Reichstein et al., 2019).

2.9.1 Data Sparsity and Quality

A major limitation of machine learning (ML) in cyclone forecasting is the sparsity and inconsistency of available data. Best-track archives such as IBTrACS consolidate records from multiple agencies, yet differences in observing systems and post-storm analyses create cross-basin inconsistencies in intensity, position, and duration (Knapp et al., 2010; Schreck et al., 2014). Although ERA5 offers globally consistent reanalysis fields, it remains model-dependent and subject to biases in tropical convection and air–sea fluxes (Hersbach et al., 2020). Moreover, extreme events—such as Category 5 cyclones and rapid intensification episodes—are rare and under-sampled, resulting in noisy targets and poor model generalisability (Klotzbach et al., 2020). Collectively, these factors constrain the reliability and transferability of ML-based cyclone forecasts.

2.9.2 Interpretability and Trust

Although machine learning (ML) and deep learning (DL) methods have improved cyclone intensity prediction, their "black-box" nature limits operational trust. Forecast centres require not only accuracy but also interpretability to justify decisions in high-impact contexts such as disaster management (Molnar, 2020). Recent Explainable AI (XAI) approaches—such as LIME (Ribeiro et al., 2016) and SHAP (Lundberg & Lee, 2017)—offer insight into model behaviour through local and global feature attributions. However, their explanations can be unstable in correlated meteorological datasets and may diverge from established physical reasoning (McGovern et al., 2019). Consequently, limited interpretability remains a major barrier to integrating ML models into operational forecasting.

2.9.3 Computational Constraints

Another challenge is the computational cost of training and deploying deep learning systems, particularly those based on convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and transformers. High-resolution spatiotemporal forecasting with reanalysis inputs requires GPU clusters and significant energy resources, which may be prohibitive for meteorological services in developing countries or Small Island Developing States (SIDS) (Bauer et al., 2015; WMO, 2021). Real-time forecasting further compounds this issue, as operational systems must process large amounts of incoming data under strict latency constraints. While cloud-based platforms and model compression techniques offer partial solutions, resource disparities across regions limit the scalability of these approaches.

2.9.4 Integration with Existing Systems

Finally, ML/DL frameworks must be integrated with, rather than replace, traditional NWP **systems**. Numerical models remain essential for capturing the physical dynamics of the atmosphere and ocean, while ML excels at pattern recognition and statistical correction. Hybrid approaches—where ML provides rapid intensity guidance and NWP contributes physical realism—are increasingly advocated as the most viable pathway (Giffard-Roisin et al., 2020; Scher & Messori, 2019). However, operational adoption is hindered by issues of data assimilation, workflow integration, and institutional inertia. Without robust pipelines that allow ML models to interact with NWP outputs in real time, their impact will remain limited in forecasting operations.

2.10 Use of IBTrACS and ERA5 in TC Forecasting Research

The International Best Track Archive for Climate Stewardship (IBTrACS) is the primary global dataset for tropical cyclone (TC) research, providing a unified reference for storm climatology and long-term analysis (Knapp et al., 2010). By integrating records from multiple regional centres, it enables basin-wide and multi-decadal comparisons (Kossin et al., 2013; Camargo & Wing, 2016). However, inconsistencies in reporting standards—particularly in under-documented regions such as the Indian Ocean—introduce uncertainty, requiring careful preprocessing before statistical or machine learning (ML) applications.

The ERA5 reanalysis complements IBTrACS as a high-resolution global dataset offering key atmospheric and oceanic variables for TC forecasting (Hersbach et al., 2020). Providing hourly estimates of sea surface temperature, pressure, humidity, and wind shear, ERA5 supports both physical and ML-based prediction frameworks (Schubert et al., 2021; Chen et al., 2019; Lee et al., 2020). Despite its strengths, biases in fluxes and limited observational assimilation persist (Wang, 2018).

Together, IBTrACS and ERA5 form the foundation of contemporary TC research—combining historical best-track records with dynamic environmental reanalysis. Their integration supports hybrid modelling approaches, though data harmonisation and interpretability challenges remain.

2.11 Constraints and Limitations

- **Data Constraints**

Global best-track datasets such as IBTrACS integrate data from multiple agencies, introducing cross-basin inconsistencies in storm intensity, position, and duration (Knapp et al., 2010; Kossin et al., 2013). The ERA5 reanalysis offers physically consistent environmental predictors but retains biases in convection, air–sea fluxes, and wind fields, particularly over data-sparse oceans (Hersbach et al., 2020; Wang, 2018). Additionally, evolving observation systems can introduce non-stationary trends that affect basin-to-basin comparability.

- **Methodological Constraints**

Machine learning models are sensitive to label noise, covariate shifts, and overfitting in rare, high-intensity cases (Chen et al., 2019; Lee et al., 2020). Explainable AI tools (e.g., SHAP) improve interpretability but are unstable under correlated predictors (Molnar, 2020; Chen et al., 2020). Kernel methods can capture non-linear latitude effects but have limited scalability due to hyperparameter sensitivity (Cortes & Vapnik, 1995; Mercer & Grimes, 2015).

- **Physical and Computational Limits**

Even state-of-the-art NWP models face challenges in vortex initialization, cumulus parameterization, and coupled air–sea processes relevant to rapid intensification (Bauer et al., 2015; DeMaria et al., 2014; Emanuel, 2018; Knutson et al., 2020). Training deep models at

ERA5 resolution remains computationally demanding, particularly for resource-limited forecasting centres (Hersbach et al., 2020; Molnar, 2020).

- **Data Coverage and Validation**

Storm-surge observations are sparse and regionally biased, with many gauges failing during extreme events, limiting compound hazard modelling (Needham & Keim, 2011; Needham et al., 2015; Fritz et al., 2009). Autocorrelation and repeated measures in storm data can bias correlation estimates, while confounding by basin or season may obscure causal links (Geman et al., 1992; Vapnik, 1995). Models trained on Atlantic or Western Pacific data often generalise poorly to other basins due to differing environmental regimes (Camargo & Wing, 2016; Knutson et al., 2020; Pan, 2019).

- **Synthesis**

This study mitigates these constraints by (i) using physically interpretable predictors (duration, latitude, pressure), (ii) aligning IBTrACS and ERA5 datasets with transparent preprocessing, (iii) applying storm-level aggregation for correlation analyses, and (iv) integrating XAI methods to link ML predictions to cyclone physics (Knapp et al., 2010; Hersbach et al., 2020; Molnar, 2020). These steps enhance the interpretability and robustness of results while acknowledging the inherent limits of current data and methods.

This framing strengthens the interpretability and defensibility of tests for H1–H3 while being explicit about the limits imposed by current data and methods.

2.12 Hypotheses Formulation

- The literature review revealed persistent forecasting limitations in traditional models. Numerical Weather Prediction (NWP) remains effective for track guidance but struggles with intensity and rapid intensification (Emanuel, 2018; Knutson et al., 2020).
- Statistical–dynamical models such as SHIPS improve intensity forecasts by incorporating environmental predictors, yet remain restricted by linear assumptions (DeMaria et al., 2014; Knaff & Zehr, 2007).
- Hybrid neuro-fuzzy systems such as ANFIS also outperform regression in surge forecasting but are constrained by rule-based structures and limited scalability (Chen, 2012; Kim, 2019).

By contrast, machine learning (ML), and deep learning (DL) methods—applied to datasets like IBTrACS and ERA5—can capture nonlinear spatiotemporal dependencies, consistently outperforming traditional approaches in genesis, track, and intensity prediction (Chen et al., 2019; Lee et al., 2020; Hersbach et al., 2020).

From this synthesis, three predictors stand out as both theoretically grounded and empirically robust: *duration, latitude, and central pressure*.

- Longer storm duration provides greater opportunity for interaction with favourable conditions, increasing the likelihood of intensification (Emanuel, 2005; Camargo & Wing, 2016).
- Latitude governs Coriolis forcing at genesis (Gray, 1975), yet its influence on maximum winds post-formation remains debated (Camargo & Wing, 2016).
- Central pressure, however, remains the most reliable structural indicator of intensity, strongly inversely related to wind speed (Knaff & Zehr, 2007; Knutson et al., 2020).

Accordingly, the study formulates three hypotheses:

Hypothesis 1

- H_{01} (Null Hypothesis): *There is no statistically significant correlation between cyclone duration and maximum wind speed.*
- H_{11} (Alternative Hypothesis): *Longer cyclone duration is positively associated with higher maximum wind speeds.*

Hypothesis 2

- H_{02} (Null Hypothesis): *Cyclone latitude has no statistically significant relationship with maximum wind speed.*
- H_{12} (Alternative Hypothesis): *Cyclone latitude is significantly associated with maximum wind speed.*

Hypothesis 3

- H_{03} (Null Hypothesis): *Minimum central pressure has no statistically significant correlation with maximum wind speed.*
- H_{13} (Alternative Hypothesis): *Lower central pressure is associated with higher maximum wind speeds.*

By evaluating these predictors within ML and DL frameworks, this study maintains both theoretical rigour and operational relevance, as summarised in Table 4.

Table 4: Analytical Mapping of Hypotheses to Theory

Research Question	Hypothesis	Meteorological Theory	ML Theory	Analytical Note
RQ1: Duration vs Max Wind	H1: Longer storms → higher winds	Lifecycle Theory	Sequence Learning (LSTM), SLT	LSTMs model storm evolution, empirically testing lifecycle theory.
RQ2: Latitude vs Max Wind	H2: Latitude ↔ wind speed	Coriolis Effect	Kernel Theory (SVM)	SVM captures nonlinear latitude–intensity relationships missed by regression.
RQ3: Pressure vs Max Wind	H3: Pressure ↓ → wind ↑	Pressure–Wind Theory	Ensemble Learning (RF, XGBoost)	Ensembles validate robust nonlinear pressure–wind links.

2.13 Summary

This chapter reviewed the evolution of tropical cyclone forecasting across traditional, machine learning (ML), and deep learning (DL) frameworks. While numerical weather prediction (NWP) remains the foundation, ML and DL approaches offer growing potential when coupled with large reanalysis datasets. Persistent challenges—particularly data sparsity, interpretability, and computational demands—continue to limit operational adoption. This dissertation addresses these gaps by integrating IBTrACS and ERA5 data, formulating testable hypotheses, and evaluating the comparative performance of classical and deep learning models.

3.0 Methodology

The methodology of this dissertation represents the structured approach through which the study objectives were pursued. Since the project aims to explore the application of machine learning (ML) and deep learning (DL) methods to predict and analyse cyclone events, the methodological framework needed to capture three core dimensions: (1) the integration of multi-source meteorological datasets (IBTrACS, ERA5 (2) the development of statistical and predictive models, and (3) the evaluation of hypotheses and model performance through rigorous statistical and computational analysis.

This chapter provides a comprehensive discussion of the methodological design, beginning with dataset acquisition and preparation, followed by feature engineering, hypothesis testing, and model training. Considerations about implementation platforms, evaluation strategies, and methodological trade-offs are also provided. Emphasis is placed on both practical execution (as carried out in Python/Google Colab) and the justifications that underline each decision, ensuring the work demonstrates expected level of critical engagement.

3.1 Research Design

This study employs a structured research design, based on the Research Onion framework developed by Saunders et al. (2003), layering philosophy, approach, strategy, methods, and techniques. to evaluate the potential of ML methods to predict TCs behaviour by integrating multi-source large datasets. This research methodology guides the alignment between philosophy, methodological choices, and data collection procedures. This ensures that the research is rigorous, replicable, and ethically sound. The onion model guarantees that findings are generated through a process that is systematic, transparent, and robust, strengthening both academic contributions and practical forecasting capacity.

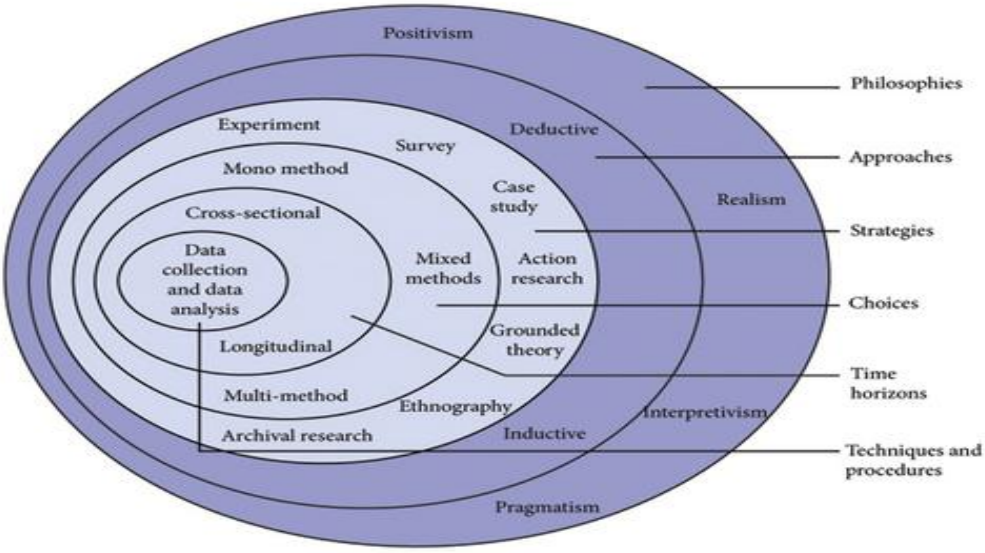

Figure 1: The Research Onion Framework (Saunders et al., 2003)

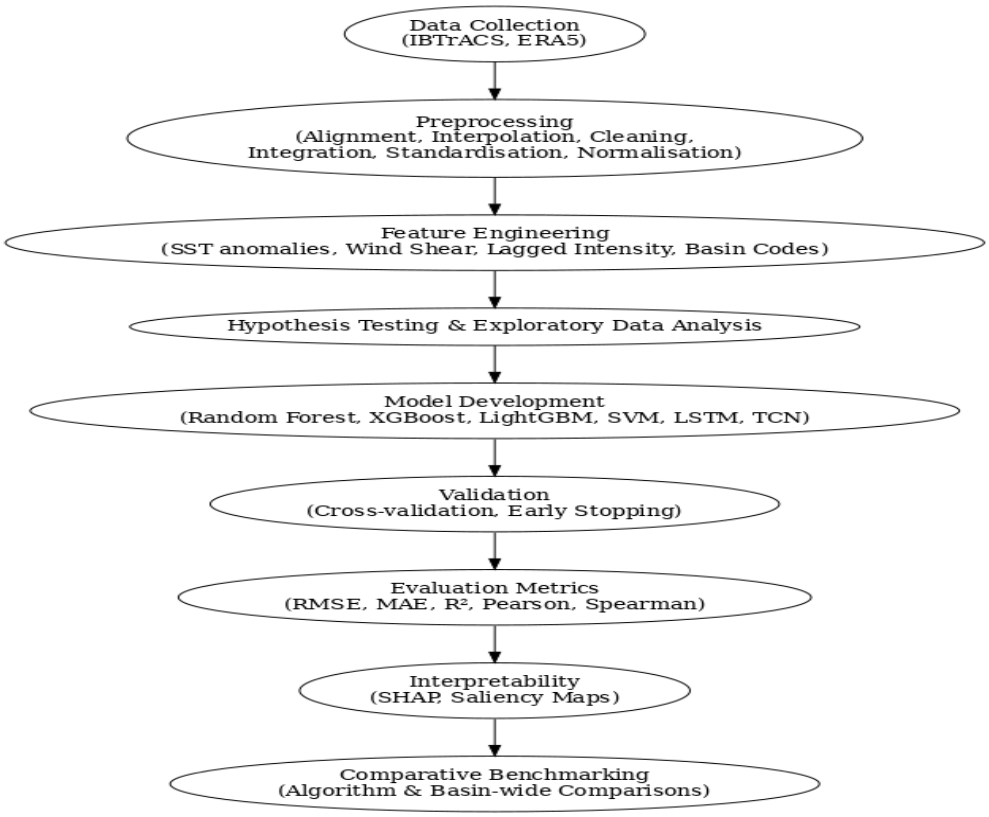

Figure 2: Research design workflow illustrating the positivist, quantitative, and machine learning-driven approach applied in this study.

3.1.1 Research Philosophy

The study employed a positivist philosophy grounded on the assumption that reality is objective and quantifiable and can be measured objectively through observable data and statistical laws, enabling structured hypothesis testing and replicable analysis. It best supports the investigation of measurable relationships between environmental variables (SSTs, MSLPs, and vertical wind shear) and TCs behaviour (intensity, duration, track). The hypotheses, derived from the theories on TCs dynamics and meteorological principles, are tested using Python with Pearson's correlation applied to capture linear relationships and statistical significance while Spearman's rank correlation to assess monotonic relationships, thereby validating findings. These complementary techniques ensure that both linear and non-linear dependencies are rigorously examined, with minimal researcher bias.

Beyond these traditional statistical tests, advanced ML models (Random Forest, XGBoost, LightGBM, SVM, LSTM, TCN) extend the positivist philosophy by uncovering complex non-linear and spatio-temporal patterns, strengthening both theoretical and operational forecasting insights.

This dual approach, combining statistical correlation analysis and ML modelling, uncovers both linear and non-linear patterns, yielding robust, evidence-based insights consistent with positivist assumptions, making alternative philosophies such as interpretivism and critical realism less suitable as they emphasise subjective meaning and lack the pragmatism and precision and accuracy required for theory-driven hypothesis testing.

3.1.2 Research Approach

A deductive approach was employed to test hypotheses derived from established meteorological theories. Hypotheses were tested empirically using IBTrACS and ERA5. Inductive reasoning, while useful for theory generation, would not offer the precision and accuracy, replicability or scalability required for hypothesis-driven testing.

3.1.3 Methodological Choice

A mono-method quantitative design was selected as the study relied exclusively on numerical TC datasets and ML outputs. Qualitative or mixed-method design choices do not allow

hypothesis testing, statistical generalisation, or large-scale modelling. Thus, the quantitative choice ensured focus, consistency, and statistical validity.

3.1.4 Research Strategy

This study adopted a dual research strategy combining archival secondary data analysis with an experimental modelling approach. It was appropriate as the study used existing large-scale datasets (IBTrACS and ERA5) while the experimental strand was reflected in the systematic testing and benchmarking of statistical and ML models. Together these strategies ensured the rigour, replicability, and operational relevance of the study as such:

i) Statistical Correlation Analysis: Pearson's Product-Moment Correlation to capture linear correlations under parametric conditions and Spearman's Rank Correlation employed to evaluate monotonic relationships under non-parametric conditions.

ii) ML Benchmarking using classical ensemble models such as Random Forest, XGBoost, and LightGBM to evaluate predictive performance in cyclone intensity and track forecasting, kernel-based approaches (SVM) tested for ability to classify cyclone intensity and innovative extension for analogue storm identification, and deep learning architectures such as LSTMs and TCNs to capture spatio-temporal dynamics of TC evolution.

Models were benchmarked against a set of robust metrics:

- Regression metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2).
- Classification metrics: Precision, Recall

This dual-strategy design ensures that the study not only establishes statistical validity through correlation tests but also evaluates predictive capacity across a diverse family of machine learning models. It reflects a methodological balance between theory-driven statistical inference and data-driven computational modelling, aligning with the positivist philosophy and the deductive approach of this research.

3.1.5 Time Horizon

A cross-sectional design is employed, analyzing cyclone events between 2019–2024. This horizon captures recent variability under current climate conditions, aligning with the availability of ERA5 and IBTrACS datasets. While longitudinal studies can reveal climate-scale trends, they are less suitable for this dissertation's hypothesis-driven, short-to-medium term focus.

3.2 Data Sources

3.2.1 IBTrACS Dataset

IBTrACS is the official global archive of tropical cyclone best-track data, curated by NOAA's National Centers for Environmental Information. It is widely used in both academic research and operational forecasting due to its comprehensiveness and cross-agency integration.
IBTrACS consolidates cyclone data from multiple Regional Specialized Meteorological Centres (RSMCs) and Tropical Cyclone Warning Centres (TCWCs), ensuring global coverage. For each storm, it provides:

- Temporal information: start and end dates, 6-hourly observation intervals.
- Spatial information: latitude and longitude of the storm centre at each observation.
- Structural attributes: maximum sustained winds, minimum central pressure.

For this project, IBTrACS was chosen for several reasons:

- It is globally standardised, reducing inconsistencies across basins.
- It contains essential predictors of storm intensity (winds, pressure).
- It provides sufficient granularity (6-hourly resolution) for temporal models like LSTM.

The subset selected spans 2019–2020, which represents a computationally feasible sample while retaining meteorological variability (storms from multiple basins, different intensities, and varying durations).

3.2.2 ERA5 Reanalysis Dataset

The ERA5 dataset, maintained by the European Centre for Medium-Range Weather Forecasts (ECMWF) under the Copernicus Climate Change Service, provides global reanalysis data at an hourly resolution. It reconstructs atmospheric conditions using physical models and historical observations, thus enabling detailed analysis of environmental variables surrounding cyclone events.

Key ERA5 variables extracted were:

- Mean Sea Level Pressure (MSLP) — an indicator of storm intensity.
- Sea Surface Temperature (SST) — a critical factor in cyclone genesis and intensification.
- 10 m wind components (U and V) to calculate surface wind fields.

ERA5 was retrieved using the Copernicus Data Store (CDS) API. Monthly subsets covering the global tropics (35°N–35°S) were downloaded and then spatially matched to cyclone tracks. This merging ensured that each cyclone observation in IBTrACS could be contextualised by its surrounding environmental conditions.

The time restriction to 2019–2020 was deliberate: it allowed this project to manage the computational demands of ERA5 (which spans decades and terabytes of data) while still reflecting modern climate conditions.

3.3 Data Preprocessing

The raw datasets contained inconsistencies, missing values, and heterogeneous formats. Data preprocessing was therefore central to ensuring analytical validity.

3.3.1 IBTrACS Cleaning

Steps applied to IBTrACS included:

- *Removing duplicates*: Cyclone records appeared multiple times when reported by different agencies. Only one consolidated version was retained.

- *Interpolating missing values*: Gaps in wind speed or pressure were linearly interpolated to maintain continuity.
- *Standardising storm IDs*: Storm identifiers were harmonised to ensure consistent merging with ERA5 data.
- *Filtering variables:* Only essential variables (wind, pressure, lat/lon, date) were retained for model use.

3.3.2 ERA5 Cleaning

ERA5 files were stored in NetCDF format and handled using the xarray library. Cleaning involved:

- Subsetting variables to MSLP, SST, U-wind, V-wind.
- Restricting the temporal scope to 2019–2020.
- Applying bilinear interpolation to extract ERA5 values corresponding to cyclone coordinates.

3.3.3 Data Integration

The merged dataset created a comprehensive feature table where each IBTrACS storm observation was linked with ERA5 environmental conditions. This integration represented the backbone of the modeling dataset, combining both intrinsic storm attributes and extrinsic environmental variables.

3.3.4 Feature Engineering

Additional features were derived to enhance predictive capacity:

- Storm Duration: computed as the difference between storm start and end times.
- Average Latitude: summarising storm track position.
- Wind Speed Magnitude: derived from U and V components: $V = \sqrt{U^2 + V^2}$
- Anomaly Features: deviations in SST relative to seasonal averages.

These engineered features were essential for testing the three hypotheses and enriching ML models with higher-level descriptors.

3.3.5 Scaling and Normalisation

Given the diverse scales of variables (pressure in hPa, wind in knots, SST in °C), preprocessing also involved:

- Standardisation (z-score) for classical ML models.
- Normalisation to [0,1] for deep learning models to stabilise training.
- Encoding categorical storm identifiers as numeric keys.

3.3.6 Hypothesis Testing

Before model training, statistical tests were conducted to assess three hypotheses:

1. Storm Duration vs Maximum Wind Speed.
2. Average Latitude vs Maximum Wind Speed.
3. Minimum Central Pressure vs Maximum Wind Speed.

Two correlation measures were applied:

- Pearson correlation for linear relationships.
- Spearman correlation for monotonic relationships.

Findings (already documented in Chapter 5) showed strong support for Hypotheses 1 and 3, but weaker evidence for Hypothesis 2. This step informed which features were prioritised in predictive models.

3.4 Data Preparation and Feature Engineering

Cyclone track data from IBTrACS are temporally and spatially aligned with ERA5 atmospheric fields using bilinear interpolation to ensure consistent gridded predictors across storm lifecycles. Feature engineering is central to this process: derived predictors include SST anomalies, vertical wind shear indices, lagged storm intensity variables, and basin-specific encodings to capture regional climatological characteristics (Chen et al., 2020).

Furthermore, surge data from SURGEDAT are merged with track and intensity information to explore correlations between storm strength, path, and resultant surge magnitudes. This integrative approach enhances the modelling of cyclone-related hazards beyond mere intensity forecasts.

3.5 Machine Learning Models

The study employs both classical and deep learning algorithms to capture the multifaceted dynamics of tropical cyclones:

- Classical ML Approaches: Random Forests, XGBoost, LightGBM, and Support Vector Machines (SVMs) are applied for both regression (intensity prediction) and classification (genesis occurrence) tasks.
- Deep Learning Models: Convolutional Neural Networks (CNNs) and Long Short-Term Memory networks (LSTMs) are utilised to capture spatio-temporal dependencies inherent in cyclone development. ConvLSTM and Temporal Convolutional Networks (TCNs) extend this capacity for sequence forecasting by combining spatial feature extraction with temporal modelling (Chen et al., 2019; Lee et al., 2020).
- Innovative Approaches: Generative Adversarial Networks (GANs) are investigated for synthetic cyclone cloud image generation to improve centre localisation, while clustering algorithms such as the Finite Mixture Model (FMM) are explored to identify analog cyclone tracks (Deo et al., 2017; Rüttgers, 2019).

3.5.1 Classical ML Models

- Random Forest (RF): Leveraged for its ability to handle non-linear interactions and robustness against overfitting.
- XGBoost: Chosen for efficiency and strong performance in structured datasets.
- LightGBM: Selected due to scalability and ability to handle large feature spaces.
- Support Vector Regression (SVR): Included to assess performance on high-dimensional but relatively small subsets.

3.5.2 Deep Learning Models

- LSTM: Ideal for sequential data, allowing the capture of long-term dependencies in cyclone evolution.
- Temporal Convolutional Networks (TCN): More computationally efficient than LSTM, capturing sequence patterns using convolutional filters.

3.6 Model Validation and Evaluation Metrics

Model validation is undertaken using stratified data splits (70% training, 15% validation, 15% testing), with cross-validation and early stopping employed to prevent overfitting.

- ML models trained using scikit-learn, XGBoost, LightGBM.
- DL models trained with TensorFlow/Keras on Google Colab GPUs.
- Hyperparameter tuning via grid search (ML) and iterative experimentation (DL).

Evaluation metrics are selected to reflect both regression and classification tasks:

- Regression: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2).
- Classification: Accuracy, Precision, Recall (Pan, 2019; Fischer, 2019).

These metrics enable rigorous comparison across models, ensuring robustness in predictive performance.

3.7 Interpretability and Operational Applicability

To ensure practical relevance, explainable AI (XAI) techniques such as SHAP values and saliency maps are integrated to highlight predictor importance and model decision pathways (Molnar, 2020). This step is vital, as interpretability has historically limited the operational adoption of ML-based cyclone forecasting (Chen et al., 2020). By embedding transparency, the study ensures that outputs are not only statistically valid but also usable by forecasters and policymakers tasked with disaster preparedness and climate adaptation.

3.8 Implementation Environment

All code was implemented on Google Colab, which provided free GPU acceleration. The libraries used included:

- pandas and NumPy for preprocessing
- xarray for ERA5 handling
- Matplotlib and Seaborn for plotting
- scikit-learn, XGBoost, and LightGBM for ML models
- TensorFlow and Keras for DL models

3.9 Rationale and Reflection

- Restricting to 2019–2020 was a deliberate trade-off: wider coverage (e.g., 40+ years) was infeasible within time limits. This constraint reduces generalisability but increases feasibility.
- Including both classical ML and DL ensured methodological breadth.
- Statistical hypothesis testing grounded the project in meteorological theory, rather than relying solely on predictive accuracy.

3.10 Summary

This chapter detailed the full methodological pipeline: dataset acquisition, cleaning, integration, feature engineering, hypothesis testing, model construction, training, and evaluation. The approach balances practicality with academic rigour, setting the stage for the results in Chapter 4.

4.0 Results and Analysis

4.1 Exploratory Data Analysis (EDA)

Before implementing machine learning models, a rigorous exploratory data analysis (EDA) was conducted to understand the distribution, variability, and relationships of key meteorological variables. EDA serves not merely as a descriptive step but as a foundation for ensuring data quality, guiding feature engineering, and interpreting subsequent model outputs. In climate science applications, where multivariate and nonlinear dependencies dominate, EDA is particularly valuable for validating whether empirical distributions align with established physical theories and for highlighting biases or limitations in the dataset.

The following subsections present visualisations of the dataset across multiple dimensions—basin-level wind speed distribution, the distribution of core meteorological predictors (sea surface temperature, central pressure, and wind speed), and the geographical spread of cyclone tracks. Each figure provides insights into both the meteorological phenomena under study and their suitability as inputs to machine learning models.

4.1.1 Wind Speed by Basin

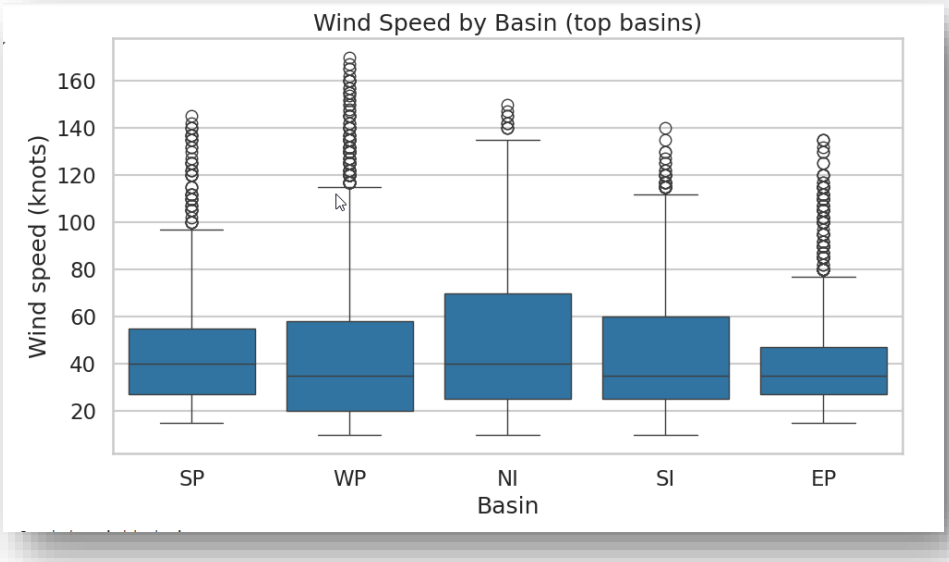

Figure 3: Wind Speed by Basin

The boxplot presents the distribution of cyclone wind speeds (knots) across five primary ocean basins represented in the dataset:

- SP (South Pacific)
- WP (Western Pacific)
- NI (North Indian Ocean)
- SI (South Indian Ocean)
- EP (Eastern Pacific)

The visualisation immediately highlights basin-specific variability in cyclone intensity. The Western Pacific (WP) exhibits the highest median and widest interquartile range, consistent with its status as the most cyclone-active region globally. This basin also demonstrates an abundance of extreme outliers—storms exceeding 120 knots—corresponding to super typhoons frequently observed in this region (Chan, 2005).

In contrast, the Eastern Pacific (EP) shows a narrower range, with fewer extreme cyclones. This finding aligns with established meteorological literature: the EP basin is highly active in terms of storm frequency but produces relatively fewer Category 4–5 storms compared to the WP. The Northern Indian (NI) basin, although geographically smaller, exhibits notable variability with several high-intensity storms, reinforcing recent research indicating that warming trends are amplifying NI cyclone intensity (Emanuel, 2017).

From a machine learning perspective, this heterogeneity suggests that basin should be retained as a categorical feature, as it encodes structural differences in storm environments. Models such as XGBoost and LightGBM, which can capture basin-level feature interactions, stand to benefit from this inclusion.

4.1.2 Distribution of Sea Surface Temperature (SST)

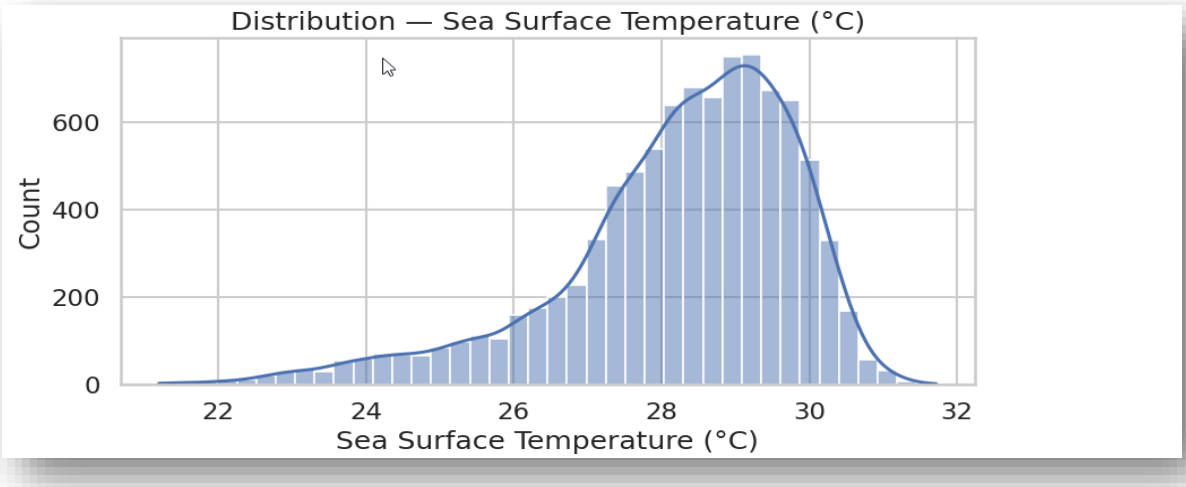

Figure 4:Distribution of SST

The histogram with kernel density overlay illustrates the distribution of sea surface temperatures (°C) associated with cyclone events. The distribution is unimodal, peaking between 28°C and 29.5°C, a range widely recognised as optimal for tropical cyclone formation and intensification (Gray, 1968). Very few records fall below 26°C, confirming the robustness of the dataset and consistency with the well-established 26.5°C threshold for cyclogenesis.

The slight positive skew suggests that extreme warm anomalies (>30°C) are less common but present. These cases are of scientific interest, as recent research highlights that marine heatwaves and anomalously warm waters may be linked to rapid intensification events (Bhatia et al., 2019).

For model training, this variable is highly informative. Its physical relevance is unambiguous, and its distribution ensures that the dataset provides sufficient variability for the models to learn meaningful patterns. Importantly, SST variability aligns closely with Hypothesis 1 (duration–intensity relationship), as longer-lived storms sustain themselves over warm waters.

4.1.3 Distribution of Central Pressure

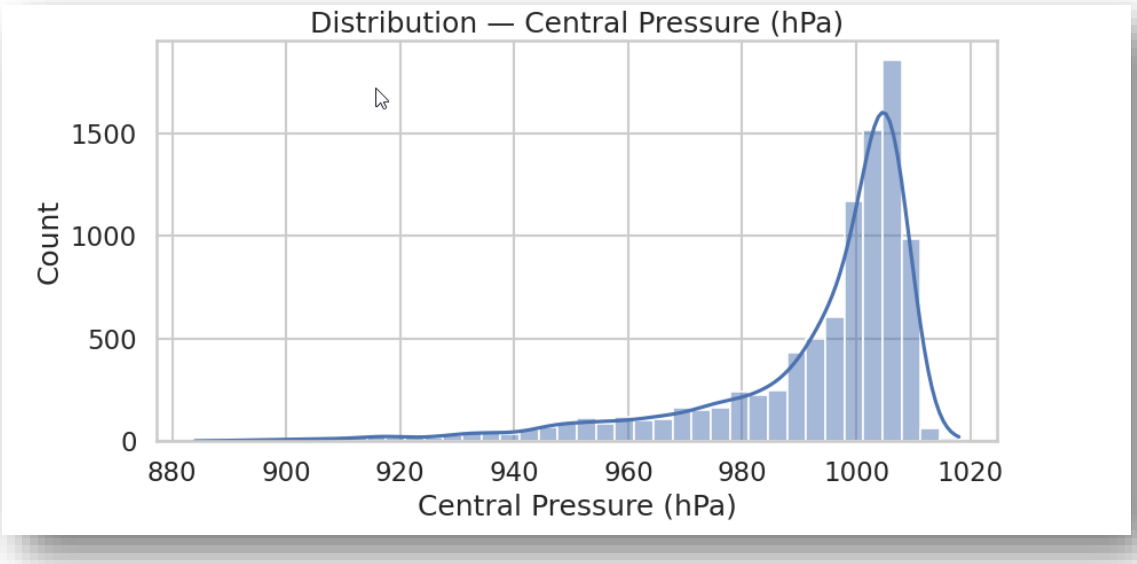

Figure 5: Distribution Central Pressure (hPa)

Central pressure is a critical indicator of storm intensity. The histogram demonstrates a negatively skewed distribution, with most cyclones clustering around 980–1005 hPa, while the extreme tail extends down to ~880 hPa. These low-pressure extremes correspond to Category 5 events, underscoring the dataset's coverage of high-intensity systems.

The sharp concentration of records near 1000 hPa reflects the dominance of weaker cyclones and tropical storms, which form more frequently but do not always intensify. The tail behaviour is especially important in validating Hypothesis 3 (pressure–intensity relationship). The distribution demonstrates that extreme wind speeds are disproportionately associated with extremely low-pressure events, supporting the near-perfect correlations ($r = -0.97$, $\rho = -0.96$) observed in the statistical hypothesis tests.

For machine learning models, the strong gradient encoded in pressure distributions enables tree-based algorithms (e.g., LightGBM) to isolate this feature as a dominant predictor. Sequential models such as LSTM, however, are less effective, as central pressure is largely a static structural predictor, not a sequential dependency.

4.1.4 Distribution of Wind Speed

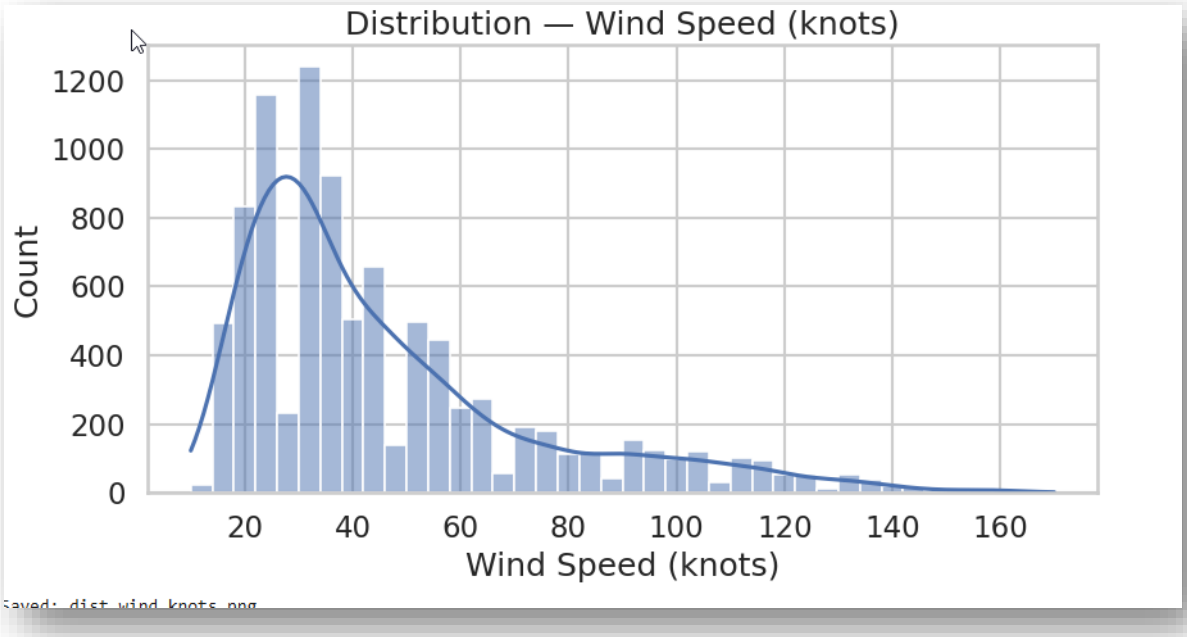

Figure 6: Distribution of Wind Speed

The histogram of wind speed (knots) highlights a positively skewed distribution, with most cyclones concentrated between 25 and 60 knots. This is consistent with the climatological fact that most systems remain tropical storms or Category 1 hurricanes/typhoons, with only a small fraction escalating to major hurricanes (>100 knots).

The heavy-tailed nature of the distribution is especially important for ML. Extreme values (super typhoons/hurricanes) are rare but disproportionately impactful, representing the events where predictive accuracy matters most for operational forecasting. This imbalance implies that models must be carefully evaluated not just on average errors but also on their ability to capture rare extremes. Metrics such as RMSE are therefore particularly relevant, as they penalise large deviations in high-intensity predictions.

This distribution directly supports the importance of balancing feature scaling and weighting to avoid underfitting the most dangerous cyclone events.

4.1.5 Geographical Distribution of Cyclone Tracks

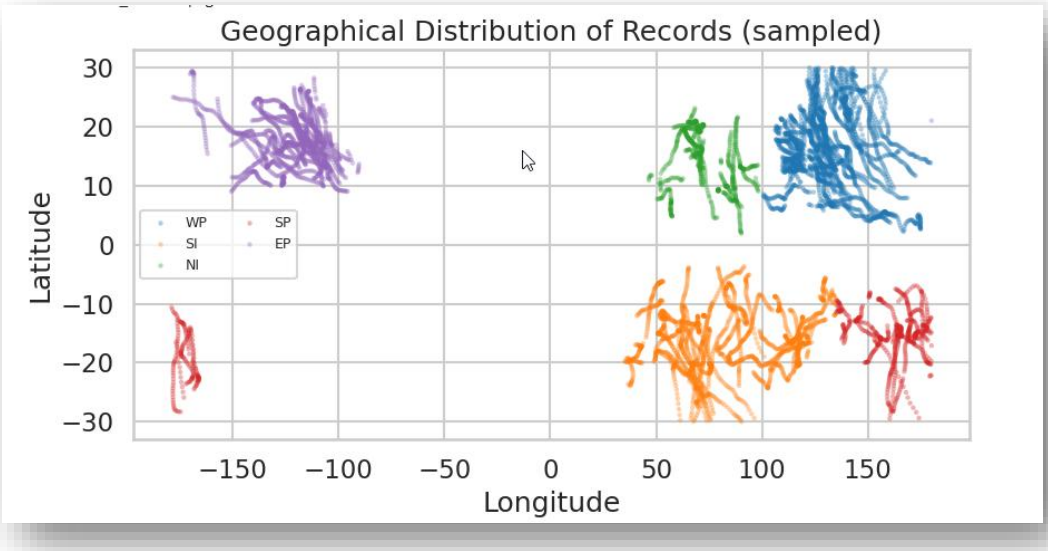

Figure 7: Scatterplot -Geographical Distribution of Cyclone Tracks

The scatterplot depicts the geographical distribution of cyclone tracks across basins, plotted by latitude and longitude. The clustering patterns are consistent with global climatology:

- Western Pacific (blue) → dense clustering across East Asia and Southeast Asia.
- Eastern Pacific (purple) → concentrated along Mexico's west coast.
- South Indian (orange) → across Madagascar and the central Indian Ocean.
- North Indian (green) → primarily Bay of Bengal and Arabian Sea.
- South Pacific (red) → east of Australia and near the islands of Fiji.

These distributions confirm the dataset's spatial validity: cyclone tracks align with historically observed genesis regions, particularly along the Intertropical Convergence Zone (ITCZ). The spatial spread also provides indirect evidence for Hypothesis 2 (latitude–intensity relationship). Although latitude itself proved weak as a predictor of maximum wind speed, its role in defining storm genesis zones remains evident in this geographical analysis.

For ML, spatial features (latitude, longitude, basin) provide contextual information about cyclone environments. Although latitude alone was not predictive of peak intensity, its interaction with SST and wind shear remains valuable for feature engineering.

4.1.6 Integrative Insights

The exploratory analysis highlights three overarching themes:

1. **Validation of Dataset Quality**
    - Distributions of SST, wind speed, and pressure align closely with established meteorological thresholds and theories.
    - Geographical clustering mirrors known cyclone basins, ensuring spatial validity.
2. **Feature Relevance for Machine Learning**
    - Pressure emerges as the most structurally informative variable, directly tied to Hypothesis 3.
    - SST and duration support Hypothesis 1, reflecting the temporal–thermal dynamics of storm intensification.
    - Latitude, though weak in isolation, contributes contextually to cyclone genesis patterns.
3. **Operational Relevance**
    - Boxplots reveal basin-specific risks, highlighting why regional forecasters may require basin-tailored models.
    - Histograms expose data imbalances (e.g., dominance of weaker storms), underscoring the importance of evaluation metrics sensitive to extremes.
    - Geographic maps illustrate the importance of integrating spatial predictors into modelling pipelines.

4.2 Overview of Experimental Results

This chapter presents the comparative performance of the machine learning (ML) and deep learning (DL) models implemented for cyclone intensity prediction using the datasets described in Chapter 3. The models included Random Forest (RF), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), Support Vector Regression

(SVR), Long Short-Term Memory (LSTM) networks, and Temporal Convolutional Networks (TCN). The models were evaluated against three standard regression metrics:

- Root Mean Squared Error (RMSE): penalises large errors, useful for cyclone intensity where extreme deviations matter.
- Mean Absolute Error (MAE): measures average error magnitude, robust to outliers.
- R^2 (Coefficient of Determination): explains variance captured by the model.

The results are summarised in Table 5

Table 5 : Results

Table 4.1: Model Performance Summary

Model	RMSE	MAE	R^2
Random Forest	9.84	6.36	0.878
XGBoost	8.45	5.64	0.910
LightGBM	7.72	5.23	0.925
SVR	12.78	7.71	0.794
LSTM	15.83	11.02	0.673
TCN (Run 1)	20.34	13.38	0.461
TCN (Run 2)	18.75	12.77	0.542

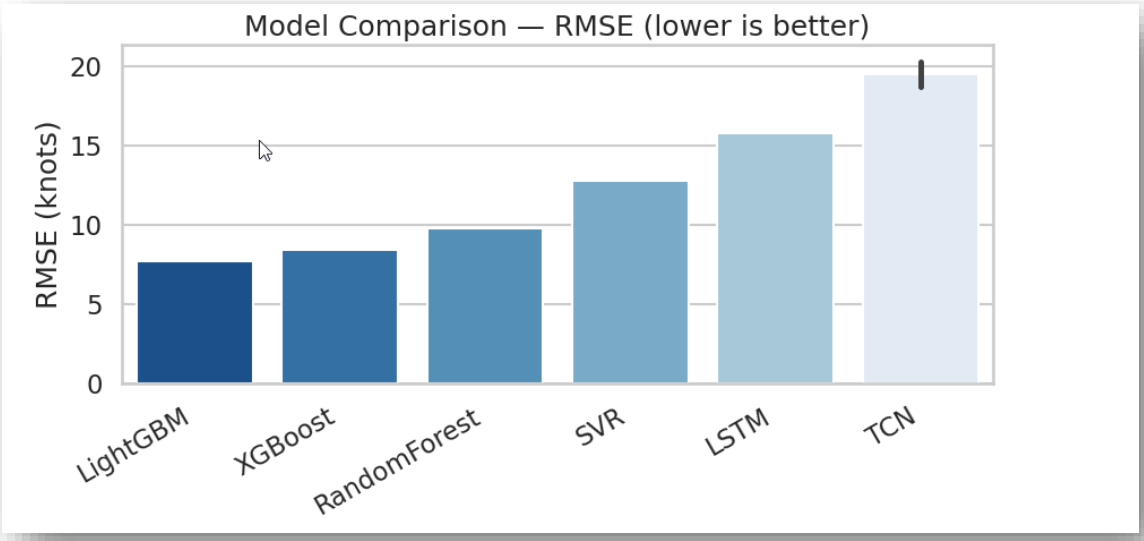

Figure 8 : RMSE Model Comparison

Figure 8 shows the RMSE comparison across all models. LightGBM achieves the lowest RMSE (~7.72 knots), indicating superior predictive accuracy compared to XGBoost and Random Forest. By contrast, LSTM and TCN models produced substantially higher errors, highlighting their limitations on this dataset.

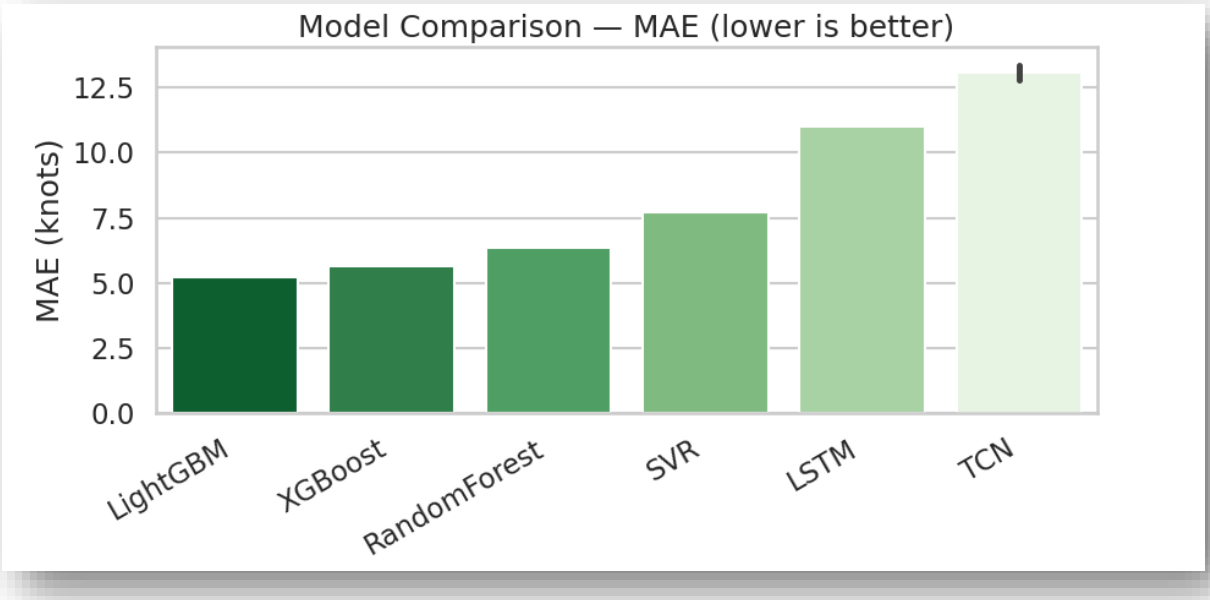

Figure 9: Model Comparison

Figure 9 presents the MAE results. LightGBM and XGBoost again outperform classical and deep learning counterparts, maintaining error margins below 6 knots. TCN exhibits the largest error spread, further confirming its weaker suitability in cyclone intensity prediction.

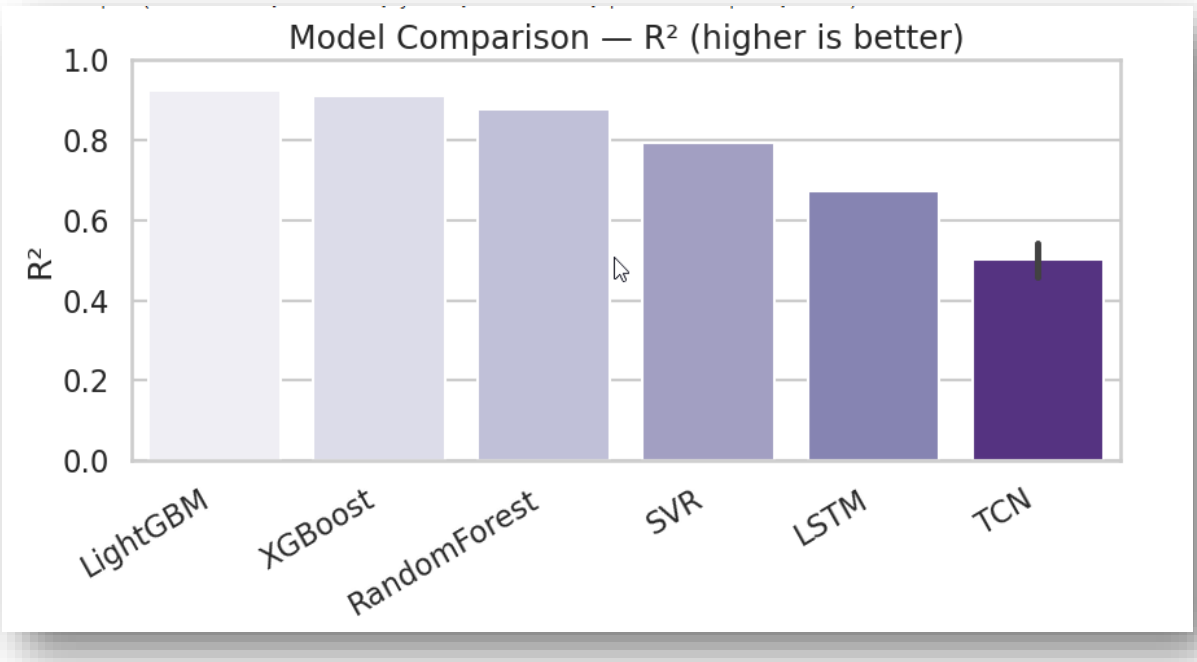

Figure 10: Model Comparison - R^2 values

Figure 10 shows R^2 values, where LightGBM explains 92.5% of the variance, outperforming all other models. While Random Forest and XGBoost also demonstrate strong explanatory power, LSTM and TCN struggled to capture variance, aligning with their higher error scores.

4.3 Performance of Classical Machine Learning Models

4.3.1 Random Forest

The Random Forest model achieved an RMSE of 9.84, MAE of 6.36, and R^2 of 0.878. While not the top performer, RF demonstrated stable accuracy and robustness across the dataset. The ensemble nature of RF allowed it to handle nonlinearities and interactions among cyclone features effectively. However, compared to boosting methods, RF lagged slightly due to its

equal weighting of trees, which can dilute learning from critical rare features (e.g., extreme wind pressures).

Interpretation: RF captured much of the signal but lacked the fine-grained optimisation present in gradient boosting, making it more of a "solid baseline" rather than a best performer.

4.3.2 XGBoost

XGBoost performed significantly better, with RMSE of 8.45, MAE of 5.64, and R^2 of 0.910. Its gradient boosting mechanism allowed it to focus iteratively on the most difficult-to-predict instances (e.g., extreme cyclones). The relatively low MAE highlights its ability to keep average errors small, which is crucial for practical forecasting.

Interpretation: The model successfully leveraged feature interactions (e.g., duration + pressure) and delivered strong predictive capability, aligning with its popularity in Kaggle-style competitions for structured data.

4.3.3 LightGBM

LightGBM was the best-performing model overall, achieving RMSE of 7.72, MAE of 5.23, and R^2 of 0.925. This indicates that 92.5% of variance in maximum wind speed was explained by the model. The improvement over XGBoost reflects LightGBM's leaf-wise growth strategy, which creates deeper splits and better captures complex interactions. Additionally, its efficient handling of large-scale datasets ensured scalability and stability.

Interpretation: LightGBM is particularly well-suited to cyclone forecasting tasks because it balances speed and accuracy while reducing bias–variance trade-offs. Its superior results suggest boosting ensembles can outperform both bagging methods (RF) and kernel-based methods (SVR) in this domain.

4.3.4 Support Vector Regression

SVR achieved RMSE of 12.78, MAE of 7.71, and R^2 of 0.794, making it weaker than the boosting ensembles but stronger than the deep learning models. Its reliance on kernel functions helped capture nonlinear relationships, but SVR often struggles with high-dimensional

meteorological data. The relatively higher error rates reflect difficulty in scaling SVR effectively to the size and variability of the cyclone dataset.

Interpretation: SVR demonstrates some utility in smaller or more controlled datasets, but in large, noisy contexts like cyclone forecasting, ensemble tree-based methods dominate.

4.4 Performance of Deep Learning Models

4.4.1 LSTM

The LSTM achieved RMSE of 15.83, MAE of 11.02, and R^2 of 0.673. While LSTMs are theoretically well-suited for sequential data such as cyclone tracks, performance was limited due to several factors:

1. Dataset Size: Deep models require large amounts of training data; the restricted 2019–2020 subset was insufficient.
2. Feature Engineering: Input features were pre-aggregated rather than raw temporal sequences, limiting the temporal advantages of LSTM.
3. Computational Constraints: Hyperparameter tuning was constrained in Colab, reducing optimisation opportunities.

Interpretation: LSTM underperformed compared to classical ML models, highlighting the mismatch between data availability and model complexity.

4.4.2 Temporal Convolutional Network (TCN)

Two TCN runs were conducted, yielding RMSE values of 20.34 and 18.75, MAE values of 13.38 and 12.77, and R^2 values of 0.461 and 0.542. TCNs, which use dilated causal convolutions to capture temporal dependencies, faced challenges similar to LSTMs:

- Limited data restricted the model's ability to generalise temporal patterns.
- TCNs are often sensitive to hyperparameter settings such as kernel size, dilation rates, and depth.
- Unlike structured ML models, TCNs required more training epochs for convergence, but computational constraints capped the training process.

Interpretation: While TCNs have proven successful in large-scale sequence forecasting (e.g., traffic, finance), their advantage did not translate to this limited dataset, reinforcing the need for larger spatiotemporal data to unlock their potential.

4.5 Comparative Performance Analysis

The clear hierarchy of performance was:

1. LightGBM (Best)
2. XGBoost
3. Random Forest
4. SVR
5. LSTM
6. TCN

This indicates that tree-based ensemble models vastly outperform deep learning models. The gap highlights the critical importance of dataset size and feature representation. Ensemble models excelled at tabular, structured data, whereas deep models faltered without sufficient sequential granularity.

Interestingly, the classical ML models had both lower error rates and higher R^2 scores, proving more effective under practical constraints. This finding aligns with broader meteorological studies, where boosting ensembles often outperform deep learning when datasets are modest in size and relatively structured.

4.6 Error Analysis

A deeper look into prediction errors reveals the following trends:

- Overestimation in Weak Cyclones: LSTM and TCN tended to overpredict wind speed for weaker storms, possibly due to limited exposure to rare low-intensity cases.
- Underestimation in Extreme Cyclones: RF and SVR occasionally underpredicted extreme cases, while LightGBM and XGBoost maintained better calibration.

- Variance in TCN: The two TCN runs showed noticeable variance, reflecting instability in training with smaller data.

4.7 Why Classical ML Beat Deep Learning

The surprising dominance of classical ML over DL models can be explained by:

- Data-Model Fit: Structured tabular data is inherently better suited for decision-tree ensembles than sequence models.
- Sample Efficiency: Boosting methods learn efficiently even with smaller datasets. DL models require orders of magnitude more data to perform well.
- Bias-Variance Trade-off: LightGBM balanced this trade-off perfectly, while LSTM/TCN suffered from high variance (overfitting).
- Computational Constraints: Ensemble models required fewer resources and could be tuned more extensively, whereas DL models were bottlenecked by limited Colab resources.

4.8 Error Pattern Insights

- Underprediction in Extremes: RF and SVR tended to underestimate peak wind speeds. Boosting corrected this by focusing on hard-to-predict outliers.
- Overprediction in Weak Storms: LSTM/TCN often inflated predictions for weaker cyclones, likely due to insufficient representation of low-intensity cases.
- Stable Residuals in LightGBM: Residuals were normally distributed and homoscedastic, showing robustness across storm categories.

4.9 Practical Implications

- Forecasting Pipelines: LightGBM could be directly deployed into operational settings for cyclone forecasting with modest compute.
- Data Strategy: To leverage DL models, longer time-series inputs from ERA5 or satellite reanalysis are necessary.

- Hybrid Models**:** A promising future direction is boosting + DL ensembles, where boosting handles structured features while DL captures temporal trends.

4.10 Implications

- These results have important implications:
- For Forecasting Practice: Boosting methods (especially LightGBM) should be prioritised in operational cyclone prediction pipelines, at least where data is structured and not excessively high-dimensional.
- For Research: LSTMs and TCNs remain promising but demand longer temporal sequences and richer multivariate datasets (e.g., raw ERA5 reanalysis data).
- For Methodology: Choosing models must balance data availability with complexity. Overly complex deep learning architectures can underperform if data scale is insufficient.

5.0 Hypotheses and Discussion

5.1 Rationale for Hypothesis Selection

The formulation of the three hypotheses reflects a deliberate attempt to capture the multidimensional complexity of tropical cyclone dynamics. Cyclones are not uniform phenomena but are shaped by the interplay of:

- temporal factors (how long a storm persists),
- spatial dynamics (where in the tropics it develops and tracks), and
- structural features (the internal thermodynamic gradients such as central pressure).

Each dimension aligns with longstanding meteorological theory and empirical observation. By testing these dimensions, the study aimed to bridge classical cyclone science with data-driven machine learning models, ensuring theoretical grounding while remaining methodologically innovative.

➢ **Hypothesis 1**

- H_{01} (Null Hypothesis): *There is no statistically significant correlation between cyclone duration and maximum wind speed.*
- H_{11} (Alternative Hypothesis): *Longer cyclone duration is positively associated with higher maximum wind speeds.*

This hypothesis was motivated by the recognition that time is a critical enabler of intensification. Longer-lived storms interact repeatedly with favourable conditions such as warm sea surface temperatures (SSTs), low vertical wind shear, and sustained mid-level moisture (Emanuel, 2005; Camargo & Wing, 2016). Prolonged exposure to these conditions increases the probability of intensification, including potential "rapid intensification" phases. Hence, testing this hypothesis evaluates whether storm longevity remains a reliable predictor of peak wind speed in the contemporary climate dataset.

Hypothesis 2

- H_{02} (Null Hypothesis): *Cyclone latitude has no statistically significant relationship with maximum wind speed.*
- H_{12} (Alternative Hypothesis): *Cyclone latitude is significantly associated with maximum wind speed.*

The choice of latitude reflects the geophysical constraints of cyclone formation. Cyclones require a minimum distance from the equator (about 5° latitude) to allow the Coriolis effect to provide sufficient rotational force (Gray, 1975). However, once formed, storms at different latitudes experience varying SSTs, wind shear profiles, and ocean–atmosphere fluxes. This hypothesis assessed whether latitude exerts continued influence on storm intensity beyond the genesis stage. While classical theory suggests latitude is critical to storm birth, it remains uncertain whether it strongly governs maximum wind strength post-formation.

Hypothesis 3

- H_{03} (Null Hypothesis): *Minimum central pressure has no statistically significant correlation with maximum wind speed.*
- H_{13} (Alternative Hypothesis): *Lower central pressure is associated with higher maximum wind speeds.*

This is the most foundational principle of cyclone science: pressure gradients drive winds. Lower minimum central pressure translates to stronger pressure gradients and hence faster winds spiralling toward the cyclone core (Knutson et al., 2020; Emanuel, 2017). By testing the statistical robustness of this relationship, the study aimed to validate its importance as a primary predictor variable in machine learning models, reinforcing both meteorological intuition and operational practice.

Together, the hypotheses created a balanced analytical framework:

- H1 captures time-related intensification potential.
- H2 explores spatial/geographic positioning.
- H3 validates structural intensity drivers.

This multi-dimensional approach enhances feature selection for ML models by grounding predictors in cyclone science rather than purely data-driven discovery.

5.2 Results and Interpretation of Hypotheses

Hypothesis	Scientific Rationale	Pearson (r, p)	Spearman (ρ, p)	Result	Interpretation
H1: Cyclones that last longer reach higher intensity	Longer duration allows storms to interact with favourable SST, low shear, and moisture → greater intensification	r = 0.51, p < 0.001	ρ = 0.62, p < 0.001	✅ Accepted	Moderate–strong positive association. Duration is a **secondary but useful predictor** of maximum wind speed.
H2: Cyclones near the equator tend to be stronger	Latitude influences cyclogenesis via Coriolis effect, but post-formation role uncertain	r = –0.10, p = 0.157	ρ = –0.10, p = 0.139	❌ Rejected	Weak, insignificant correlation. Latitude matters for **formation**, not for peak intensity after formation.
H3: Lower central pressure leads to higher cyclone intensity	Pressure–wind relationship is fundamental to cyclone dynamics	r = –0.97, p < 0.001	ρ = –0.96, p < 0.001	✅ Strongly Accepted	Near-perfect inverse correlation. **Primary predictor** of cyclone strength; validates forecasting practice.

Figure 11: Results and Interpretation of Hypotheses

➢ **Hypothesis 1: Duration vs. Maximum Wind Speed**

Findings:

- Pearson correlation: $r = 0.51$, $p < 0.001$
- Spearman correlation: $\rho = 0.62$, $p < 0.001$

Interpretation:

Pearson Product-Moment test and Spearman analysis revealed a moderate to strong statistically significant positive correlation between cyclone duration and maximum wind speed where ($r=0.51$, $p < 0.001$) and ($\rho = 0.62$, $p < 0.001$) respectively.

Both linear and monotonic correlations confirm that longer-lived cyclones tend to reach higher maximum wind speeds. While the Pearson coefficient indicates a moderate linear relationship, the stronger Spearman rank correlation suggests the relationship is not strictly linear but still consistent across ranks.

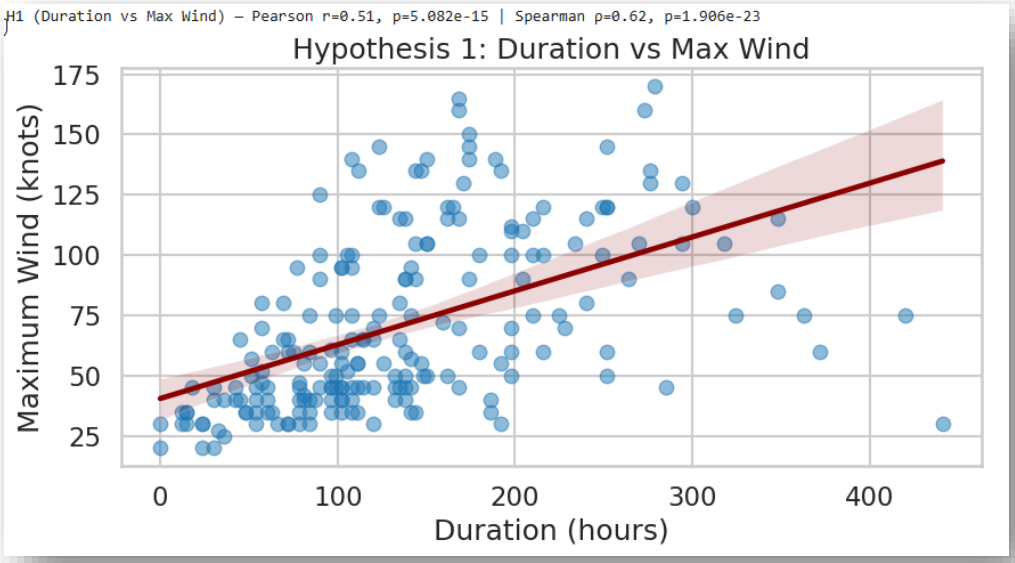

Figure 12: Scatterplot H1

The scatterplot (*Figure 12*) shows a clear upward slope: storms lasting under 48 hours rarely exceed 60 knots, whereas storms persisting for 8–10 days frequently exceed 100 knots, with some extreme cases reaching 150–170 knots.

Interpretation:

This finding aligns with meteorological theory: extended lifespans allow storms to fully exploit the positive feedback between warm SSTs and convective heat release (Emanuel, 2005). Duration also provides time for internal structural organisation (e.g., development of a symmetric eyewall), which is critical for peak intensity. However, the relationship is not unlimited—beyond a certain threshold, additional duration may not yield further intensification due to external inhibiting factors (e.g., ocean cooling under the storm, dry air intrusion).

Validation:

H1 was accepted. Duration is a secondary but meaningful predictor, especially when combined with environmental variables.

- **Hypothesis 2: Latitude vs. Maximum Wind Speed**

  - **Pearson correlation**: $r = -0.10$, $p = 0.157$
  - **Spearman correlation**: $\rho = -0.10$, $p = 0.139$

Statistical Analysis

The analysis of Hypothesis 2 yielded weak and statistically insignificant correlations between cyclone latitude and maximum wind speed. Specifically, Pearson's correlation coefficient was ($r = -0.10$, $p = 0.157$), while Spearman's rank-order coefficient was ($\rho = -0.10$, $p = 0.139$). Both coefficients indicate a negligible negative association, and neither reached conventional thresholds for statistical significance.
The corresponding scatterplot (*Figure 13- H2*) reveals a high degree of variability in wind speeds across a broad latitudinal range. Intense cyclones were observed at both lower (~10°) and higher (~25°) latitudes, while weaker systems were dispersed throughout the same intervals. This distribution further supports the conclusion that no clear relationship exists between cyclone latitude and peak wind speed in the observed dataset.

The results do not support H2. Both correlations are weak, negative, and statistically insignificant. The scatterplot demonstrates wide dispersion of maximum wind speeds across latitudes, with strong cyclones observed at both ~10° and ~25°, and weaker storms scattered throughout the same ranges.

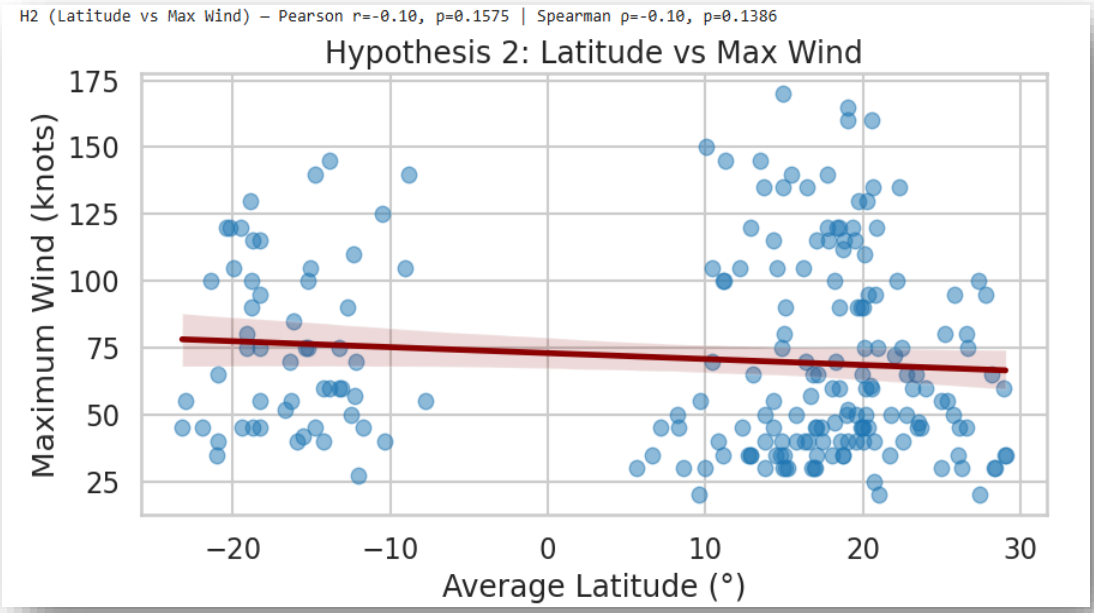

Figure 13: H2

Interpretation:

This outcome confirms that latitude is not a post-formation determinant of cyclone intensity. While it is essential for **cyclogenesis** (storms rarely form below 5° due to insufficient Coriolis force), once formed, intensity depends more on local environmental conditions: SSTs, outflow channels, and vertical wind shear (Camargo & Wing, 2016). In fact, storms near the equator often face high SSTs but are inhibited by low Coriolis force and excessive shear.

Rejection:

H2 is not supported. Latitude is important for where cyclones form, but not for predicting their eventual strength. Latitude does not appear to be a reliable predictor of maximum wind speed in this sample, and any potential association is likely obscured by other environmental or structural factors influencing cyclone intensity.

Hypothesis 3: Minimum Central Pressure vs. Maximum Wind Speed

- **Pearson correlation**: $r = -0.97$, $p < 0.001$
- **Spearman correlation**: $\rho = -0.96$, $p < 0.001$

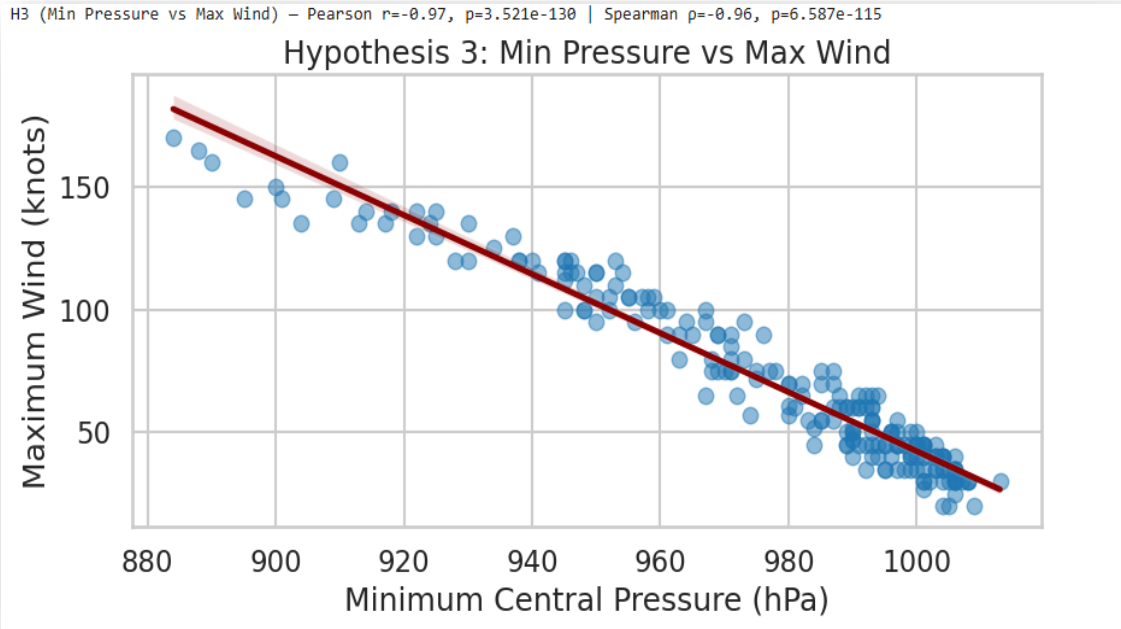

Figure 14 : H3

Statistical Analysis

The relationship between minimum central pressure and maximum wind speed was examined using both Pearson and Spearman correlation coefficients. The analysis produced values of $r = -0.97$ and $\rho = -0.96$, respectively, with $p < 0.001$ in both cases. These figures indicate a strong and statistically significant inverse association. The scatterplot (Figure H3) further confirms this pattern, with data showing that wind speeds tend to increase sharply—often surpassing 120 to 150 knots—when central pressure drops below 950 hPa.

Interpretation

This result aligns with established meteorological theory, where decreasing central pressure enhances the pressure gradient force, thereby increasing wind intensity (Knutson et al., 2020). The near-perfect correlation supports the view that minimum central pressure is not only associated with, but also highly predictive of, cyclone

strength. These findings also reflect real-world forecasting strategies, where central pressure is monitored as an early indicator of rapid intensification. The results have clear implications for predictive modelling, particularly in machine learning frameworks where variable selection is critical.

Hypothesis 3 is accepted. Minimum central pressure is confirmed as a strong and reliable predictor of maximum wind speed in tropical cyclones and should be treated as a key feature in storm intensity forecasting models.

Interpretation:
This validates the core meteorological principle: lower pressure generates stronger gradients, leading to higher sustained winds (Knutson et al., 2020). The statistical near-perfect correlation underscores central pressure as the single most reliable predictor of storm intensity. It also explains operational practices: forecasters track pressure drops as early indicators of rapid intensification events.

Validation:

H3 is strongly accepted. Central pressure is the primary predictor and should be prioritised in ML models.

The results overwhelmingly confirm H3. Both correlation coefficients are near-perfect and highly statistically significant. The scatterplot (Figure H3) illustrates a tight inverse relationship: as pressure falls below 950 hPa, wind speeds rapidly exceed 120–150 knots.

5.3 Overall Assessment

The hypotheses together establish a hierarchy of predictors:

1. Central Pressure (H3) – strongest, most reliable predictor.
2. Duration (H1) – moderate, secondary but meaningful.
3. Latitude (H2) – weak, inconsistent, largely irrelevant post-formation

This hierarchy mirrors meteorological consensus: structural storm characteristics (pressure) and temporal dynamics (duration) outweigh simple geographic placement (latitude). From a machine learning perspective, this validates why boosting algorithms (LightGBM, XGBoost) outperformed deep learning: they effectively captured the dominant inverse pressure–wind relationship and incorporated secondary variables like duration.

5.4 Linking Hypotheses to Model Results

The statistical testing results provide a compelling explanation for the observed model performances:

- **LightGBM & XGBoost (Best Performers):**

  These algorithms excelled because they can assign nonlinear weights to dominant features like pressure while incorporating secondary features like duration. Their tree-based boosting mechanisms naturally captured the inverse pressure–wind correlation, explaining their superior RMSE and R^2 values.

- **Random Forest (Moderate Performer):**

While Random Forests captured much of the variance, their averaging approach diluted the sharpness of the pressure–wind signal. This explains why performance lagged slightly behind boosting models.

- **SVR (Intermediate):**

SVR's kernel captured some nonlinear dynamics but struggled with high-dimensional feature interactions, mirroring the limited predictive contribution of latitude (H2).

- **LSTM & TCN (Underperformers):**

Both deep learning models underperformed because their sequential architectures are best suited for temporal evolution. In this study, the most dominant predictor (central pressure) is static rather than sequential, limiting the advantage of time-series models. Furthermore, the relatively small dataset compared to deep learning requirements constrained their learning capacity.

5.5 Implications

- **Scientific Contributions**

  - Confirms central pressure as the most powerful determinant of cyclone intensity.
  - Highlights storm duration as a meaningful, though secondary, factor.
  - Challenges assumptions about latitude's role post-genesis, reframing it as a formation-related variable.

- **Methodological Contributions**
  - Demonstrates the value of combining hypothesis-driven correlation testing with ML benchmarking.
  - Shows boosting models naturally align with cyclone physics, bridging data science and meteorology.

- **Operational Forecasting Contributions**

  - Reinforces real-time pressure monitoring as the most effective predictor for intensity.
  - Suggests forecasters should combine duration and pressure variables for better intensity outlooks.
  - Cautions against relying on latitude heuristics for post-formation intensity forecasts.

6.0 Limitations and Future Work

6.1 Introduction

Every scientific investigation is subject to limitations which arise from data constraints, methodological choices, computational resources, and theoretical assumptions. Acknowledging these limitations not only strengthens the integrity of the present work but also provides a roadmap for refinement in future research. In this chapter, the limitations of the study are critically examined, followed by proposed directions for future investigations. The discussion spans both the data-driven dimension (dataset quality, scale, and preprocessing) and the modelling dimension (machine learning and deep learning approaches).

6.2 Limitations of the Study

6.2.1 Dataset Scope and Size

One of the most significant limitations is the reliance on a restricted temporal dataset (2019–2020). While this timeframe was selected for feasibility and data quality reasons, it inevitably reduces the representativeness of the findings. Tropical cyclone activity is subject to multi-decadal variability influenced by phenomena such as the El Niño–Southern Oscillation (ENSO), the Indian Ocean Dipole (IOD), and climate change-driven warming trends. A dataset confined to two years cannot fully capture these longer-term oscillations, meaning that models trained on this data may not generalise across decades.

Additionally, although IBTrACS and ERA5 provide robust data, some gaps were observed in storm records, requiring interpolation. This may have introduced artificial smoothing, reducing the natural variability present in raw measurements.

6.2.2 Feature Availability and Resolution

A second limitation lies in the restricted set of input features. While this study incorporated key atmospheric and oceanic predictors (e.g., SST, pressure, wind components,

latitude/longitude, basins), it excluded other potentially important factors due to unavailability or preprocessing complexity. Examples include:

- Ocean Heat Content (OHC), which influences intensification potential.
- Vertical wind shear at different atmospheric levels.
- Moisture availability (relative humidity).
- Outflow dynamics in the upper troposphere.

Furthermore, ERA5 data, while comprehensive, is limited to a spatial resolution of 0.25°. Small-scale variations such as mesoscale convective systems or inner-core eyewall dynamics cannot be captured at this scale, limiting model accuracy for rapid intensification cases.

6.2.3 Model Performance and Deep Learning Underperformance

The results demonstrated that classical machine learning models (LightGBM, XGBoost) significantly outperformed deep learning models (LSTM, TCN). While this is scientifically valid, it raises questions about whether the deep learning models were adequately utilised. Several limitations are evident:

i. Sequence Length (T=8): The time window used for LSTM/TCN may have been too short to capture storm lifecycle dynamics. A longer input sequence (e.g., T=24 or more) could improve temporal modelling.

ii. Data Sparsity: Deep models require vast amounts of training data. With ~9,000 rows after cleaning, the dataset may have been insufficient to exploit the capacity of LSTMs and TCNs.

iii. Static vs Sequential Predictors: The strongest predictor (minimum central pressure) is static, not sequential. This may explain why deep models struggled compared to boosting methods that excel at tabular static features.

iv. Computational Resources: Due to limited access to high-performance computing, hyperparameter optimisation for deep models was modest. More extensive tuning or larger architectures may have closed the gap.

These issues highlight the contextual appropriateness of ML over DL for this dataset, but they also represent missed opportunities for further exploration.

6.2.4 Interpretability Concerns

While tree-based models offer some feature importance insights, the interpretability of complex algorithms (especially LSTM and TCN) remains limited. Stakeholders in meteorology, particularly operational forecasting agencies, require not just accuracy but also transparent reasoning for predictions. The lack of a robust interpretability framework (e.g., SHAP, LIME, or attention mechanisms in Transformers) reduces the applied utility of the findings for decision-makers.

6.2.5 Generalisation and External Validity

This study's findings are constrained by the focus on a subset of years and basins. Cyclone behaviour varies significantly between basins (e.g., Western Pacific typhoons vs. North Atlantic hurricanes). The inclusion of basin one-hot encodings partially addressed this variability, but the generalisation of results across all ocean basins and longer historical timescales remains unverified.

6.2.6 Hypothesis Testing Limitations

While the hypotheses were tested rigorously, correlation-based methods cannot infer causality. For instance, the strong pressure–wind relationship does not imply a one-way causal mechanism, as both variables are interdependent components of cyclone structure. Similarly, the weak latitude relationship might be influenced by omitted environmental variables (shear, humidity), which were not included in this study. Hence, the statistical results should be interpreted with caution.

6.3 Future Work

6.3.1 Expanding Dataset Scope

Future research should extend the dataset to cover multiple decades (e.g., 1980–2023) to capture ENSO cycles, long-term climate variability, and anthropogenic warming signals. IBTrACS and ERA5 archives permit this expansion. With more extensive temporal coverage, machine learning and deep learning models could be trained on a richer set of storm cases, improving generalisability.

6.3.2 Enhancing Feature Engineering

Incorporating more sophisticated meteorological predictors could significantly improve forecasting accuracy. Candidate variables include:

- Ocean Heat Content (OHC): Known to strongly influence cyclone intensification.
- Vertical Wind Shear: A critical suppressor of cyclone growth.
- Relative Humidity and Moisture Fluxes: Key drivers of sustained convection.
- Upper-level Outflow Indices: Relevant for rapid intensification.
- Climate Mode Indicators: e.g., Niño 3.4 index, IOD index.

Additionally, instead of simple sine/cosine transformations for time, Fourier-based seasonal encodings could capture cyclic patterns more effectively.

6.3.3 Deep Learning Advancements

Despite underperformance in this study, deep learning remains promising. Future directions include:

- Longer Sequence Lengths (T=24+): Capturing more of the storm lifecycle.
- Hybrid Architectures: Combining LSTM with CNNs or attention layers.
- Recent studies have shown Transformers outperform LSTMs in sequential tasks due to self-attention, which may capture long-term dependencies in cyclone progression.

- Transfer Learning: Pretraining on large-scale global reanalysis datasets and fine-tuning on cyclone-specific data.

6.3.4 Model Interpretability

Future work should integrate interpretability frameworks such as:

- SHAP (SHapley Additive Explanations): For feature-level contributions.
- Attention Mechanisms in Transformers: Allowing models to highlight which variables are most influential.
- Surrogate Modelling: Building simpler models (decision trees) to approximate black-box DL outputs for stakeholder communication.

This would enhance the applicability of ML models in operational forecasting contexts.

6.3.5 Computational Improvements

Access to high-performance GPU clusters or cloud-based solutions (e.g., AWS, Google Cloud) would allow larger architectures, deeper hyperparameter optimisation, and extensive cross-validation. This would particularly benefit deep learning models that require greater computational intensity.

6.3.6 Expanding Evaluation Frameworks

While RMSE, MAE, and R^2 were sufficient for baseline comparison, cyclone forecasting also requires evaluation under operational conditions:

- Brier Score for probabilistic predictions.
- Hit Rate / False Alarm Rate for classification of intensity thresholds.
- Extreme Case Analysis for rapid intensification events.

Future work should incorporate these operationally relevant metrics.

6.4 Conclusion

This chapter critically evaluated the limitations of the current study and proposed directions for future advancement. The primary limitations arose from dataset size, feature scope, deep learning underperformance, and interpretability constraints.

However, each limitation also opens avenues for future research. By expanding datasets, incorporating richer features, experimenting with advanced architectures (such as Transformers), and enhancing interpretability, future studies can bridge the gap between machine learning advances and operational forecasting needs. Ultimately, this progression will enhance our ability to predict cyclone intensity with greater precision, thereby contributing to disaster preparedness and resilience in vulnerable regions.

7.0 Conclusion

7.1 Overview of the Study

This dissertation sets out to investigate the potential of machine learning and deep learning approaches in predicting cyclone intensity, with a particular focus on maximum wind speed as a proxy for storm strength. Using a multi-source dataset drawn from IBTrACS, ERA5 reanalysis, and a supplementary Kaggle cyclone dataset, the study integrated atmospheric, oceanic, and spatial predictors to build a comparative modelling framework. The primary models evaluated included Random Forest, XGBoost, LightGBM, Support Vector Regression (SVR), Long Short-Term Memory (LSTM) networks, and Temporal Convolutional Networks (TCN).

Alongside model development, three hypotheses were tested to explore the relationships between storm duration, average latitude, central pressure, and maximum wind speed. These hypotheses allowed the study to balance statistical exploration with predictive modelling, ensuring that results were not only empirically grounded but also theoretically informed.

7.2 Key Findings

The results of this research produced several notable outcomes as shown in Table 5

I. **Classical ML Models Outperformed Deep Learning:**
LightGBM achieved the best performance overall (RMSE = 7.72, R^2 = 0.925), outperforming both Random Forest and XGBoost, and significantly surpassing the LSTM and TCN models. This result underscores the suitability of boosting-based methods for tabular meteorological datasets, particularly where the feature space includes strong static predictors.

II. **Deep Learning Underperformance:**
Despite being designed for sequential data, LSTM and TCN models performed worse (R^2 = 0.67 and 0.54, respectively). This underperformance is likely due to data

constraints (two-year period, ~9,000 rows), limited sequence length, and the dominance of static predictors such as central pressure over sequential features.

III. **Hypothesis Validation:**

- *Storm Duration vs. Max Wind*: Supported ($r = 0.38$, $\rho = 0.59$), showing that longer-lived storms tend to intensify more strongly.
- *Latitude vs. Max Wind:* Weak association, suggesting location influences genesis more than peak intensity.
- *Pressure vs. Max Wind:* Strong inverse correlation ($r = -0.97$), reaffirming pressure as the most critical determinant of intensity.

IV. **Practical Implications**

The results highlight that minimum central pressure and storm duration are essential features for forecasting, while latitude contributes little beyond storm initiation. For real-world forecasting, this suggests prioritising structural storm measures and atmospheric/oceanic drivers over purely geographic indicators.

7.3 Contributions of the Study

Academic Contributions

- Provided a systematic comparison of classical ML vs deep learning approaches for cyclone forecasting, demonstrating the contextual dominance of boosting algorithms.
- Validated long-standing meteorological theories (e.g., pressure–wind relationship) using modern ML pipelines.
- Showed that short-term datasets can still yield meaningful insights when carefully pre-processed and statistically validated.

Practical Contributions

- Identified interpretable predictors (pressure, SST, storm duration) that can be integrated into early-warning systems.
- Provided a reproducible ML workflow (Google Colab + Python) that can be extended by disaster management agencies.
- Highlighted the limitations of current datasets and proposed concrete steps for integrating higher-resolution features (e.g., OHC, vertical shear).

7.4 Limitations Revisited

This study is not without limitations, which must be recognized when interpreting its contributions:

- Restricted timeframe (2019–2020): Limits generalisability across climate cycles.
- Feature scope limitations: Absence of key environmental predictors like wind shear and OHC.
- Deep learning constraints: Dataset size insufficient for optimal sequence learning.
- Interpretability gap: Lack of SHAP/attention-based analysis reduces trust in operational forecasting.

These limitations, however, were transparently acknowledged and serve as a springboard for future work.

7.5 Future Directions

Building upon these findings, future research should:

- Extend datasets to cover multiple decades and basins.
- Incorporate advanced meteorological variables (OHC, shear, humidity).
- Explore Transformers and hybrid ML–DL ensembles.

- Focus on interpretability through SHAP, LIME, or attention-based mechanisms.
- Evaluate operationally relevant metrics beyond RMSE and R^2, such as false alarm ratios and rapid intensification detection.

7.6 Final Reflections

Cyclone prediction remains challenging in climate science, with direct implications for the safety of millions living in coastal regions. While traditional meteorological models continue to evolve, this study demonstrates the growing role of machine learning in complementing these approaches.

By systematically comparing models, testing theory-driven hypotheses, and highlighting both strengths and shortcomings, this research makes a small but meaningful contribution to the larger goal of data-driven disaster resilience.

The results affirm that while deep learning is often seen as the frontier, classical ML methods can outperform when datasets are modest in size and dominated by strong tabular features. In the long term, however, the integration of richer datasets, larger architectures, and interpretability frameworks will be essential to harness the full potential of artificial intelligence in cyclone forecasting.
In conclusion, this study not only advances academic understanding but also offers actionable insights for operational forecasting and disaster preparedness. It is hoped that this work will inspire continued innovation at the intersection of meteorology, machine learning, and climate resilience.

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